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Modulation of the gut microbiome to treat mental disorders or diseases of the central nervous system

Abstract

The present disclosure relates to methods of treating at least one symptom of a mental disorder or disease of the central nervous system in a subject by modulating the amount of GABA produced in the subject's gut. The present disclosure also relates to methods of culturing the bacterial strain new bacterial strains. Also disclosed are methods of identifying bacterial strains capable of producing GABA, and engineering strains to produce GABA.

Inventors:	Strandwitz; Philip (Medford, MA), Lewis; Kim (Newton, MA)
Applicant:	Holobiome, Inc. (South Boston, MA)
Family ID:	1000008778854
Assignee:	Holobiome, Inc. (Boston, MA)
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Primary Examiner: Davis; Ruth A

Attorney, Agent or Firm: Fenwick & West, LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/084,511 filed on Sep. 12, 2018, now U.S. Pat. No. 11,116,804, issued on Sep. 14, 2021, which is a National Application under 35 USC § 371 of International PCT Application PCT/US2017/022091, filed on Mar. 13, 2017, which claims priority to, and benefit of, U.S. Provisional Application No. 62/307,991, filed Mar. 14, 2016, the contents of each of which are herein incorporated by reference in their entireties.

INCORPORATION OF SEQUENCE LISTING

(1) The contents of the text file named “HOBE-001-02US_SeqList.txt,” which was created on Aug. 6, 2021 and is 7.6 MB in size, are hereby incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

(2) The present disclosure relates to compositions and methods for treating at least one symptom of

a disease in a subject. In some cases, the disease is a mental disorder or a disease of the central nervous system. The present disclosure teaches treatment of the disease by modulating (e.g., increasing) the amount of endogenous GABA in a subject's body. In some embodiments, the present disclosure teaches modulating (e.g., increasing) the amount of GABA produced in the subject's gut by bacteria in the gut. For example, the present disclosure teaches administration to a subject in need thereof bacteria that are capable of producing GABA (e.g., inside a human gut).

(3) The present disclosure also relates to methods of culturing previously uncultured bacterial strains. For instance, the present disclosure teaches the previously uncultured bacterial strain *Evtepia gabavorous* KLE1738. As set forth herein, newly uncultured bacterial strains such as *Evtepia gabavorous* KLE1738 can be cultured by providing growth factors necessary for the bacteria's growth and reproduction.

(4) Also disclosed are methods of identifying bacterial strains capable of producing certain growth factors. For instance, described herein are methods of identifying bacterial strains capable of producing GABA, for instance under physiologically relevant conditions such as at a physiologically relevant pH.

BACKGROUND

(5) The gut microbiome affects certain gastrointestinal and metabolic disorders, such as irritable bowel syndrome (IBS), Crohn's disease, ulcerative colitis, celiac disease, obesity, heart disease, type I and II diabetes, and colon cancer.

(6) Microbiological studies have so far been limited, by necessity, to cultivable microorganisms. By some estimates, in external environments, 99% of bacteria are thought to be uncultured. The development of new techniques for culturing previously uncultured or unculturable bacteria can thus help to expand the scope of microbiology research.

SUMMARY OF THE INVENTION

(7) The present disclosure provides compositions and methods for treating diseases such as mental illness or diseases of the central nervous system. In some embodiments, the present disclosure teaches therapeutic compositions comprising one or more bacteria (e.g., purified bacteria) that are capable of producing GABA. The bacteria can be capable of producing GABA under physiologically relevant conditions, including within a human gut. The present disclosure also provides methods of treating a subject in need thereof comprising administering to a subject a therapeutic composition comprising GABA-producing bacteria. As set forth herein, the GABA-producing bacteria can produce GABA in the subject's gut. The GABA can diffuse into other systems of the subject's body (e.g., the circulatory and nervous systems). There, the endogenous GABA can act as a neurotransmitter. In some embodiments, increased levels of GABA (e.g., in the nervous system) can improve the symptoms of the mental illness or disease of the central nervous system.

(8) In some embodiments, the present disclosure also provides methods for identifying bacteria that produce GABA in humans at a physiologically relevant pH range and uses for these bacteria to modulate GABA levels in humans to treat mental illness.

(9) The present disclosure also relates to a method of culturing previously uncultivated bacterial species. For instance the present disclosure teaches the isolation and characterization of a bacterial species KLE1738, provisionally named *Evtepia gabavorous*. Growth of *E. gabavorous* requires the presence of the growth factor GABA, which can be supplied by GABA-producing bacteria such as *Bacteroides fragilis* KLE1758.

(10) In one aspect, the present disclosure provides a therapeutic composition comprising at least one purified bacterial population consisting of bacteria capable of producing GABA in a subject in need thereof.

(11) In some embodiments, the at least one bacterial population consists of a bacteria comprising a 16S rDNA sequence at least about 95% identical to a 16S rDNA sequence selected from one of Seq. ID. Nos. 1-31 set forth in Table 1. In some embodiments, the at least one purified bacterial

population consists of bacteria selected from the group consisting of: *Bacteroides caccae* KLE1911; *Bacteroides clarus* KLE1930; *Bacteroides dorei* KLE1912; *Bacteroides finegoldii* KLE1931; *Bacteroides fragilis* KLE1958; *Bacteroides massiliensis* KLE1932; *Bacteroides ovatus* KLE1770; *Bacteroides stercoris* KLE1933; *Bacteroides thetaiotaomicron* KLE1934; *Bacteroides uniformis* KLE1913; *Bacteroides vulgatus* KLE1910; *Bacteroides xylanisolvens* KLE1935; *Bifidobacterium adolescentis* KLE 1879; *Blautia obeum* KLE1914; *Blautia wexlerae* KLE1916; *Butyricimonas virosa* KLE1938; *Clostridium perfringens* KLE1937; *Clostridium sordellii* KLE1939; *Clostridium* sp. KLE1862; *Clostridium* sp. KLE1918; *Coprobacillus* sp. KLE1779; *Coproccoccus* sp. KLE1880; *Dorea longicatena* KLE1917; *Eggerthella lenta* KLE1926; *Eubacterium rectale* KLE1922; *Gordonibacter pamelaee* KLE1915; *Oscillibacter* sp. KLE1928; *Parabacteroides distasonis* KLE2020; *Parabacteroides merdae* KLE1863; *Ruminococcus gnavus* KLE1940; *Turicibacter sanguinis* KLE1941, and combinations thereof.

(12) In some embodiments, the at least one purified bacterial population consists of a bacteria comprising a 16S rDNA sequence at least about 95% identical to a 16S rDNA sequence selected from one of Seq. ID. Nos. 32-274 set forth in Table 2. In some embodiments, the at least one purified bacterial population consists of bacteria comprising a 16S rDNA sequence having at least 95% similarity to the 16S rDNA sequence selected from one of Seq. ID. Nos. 305-2217 set forth in Table 10. In some embodiments, the at least one purified bacterial population consists of bacteria comprising a DNA sequence which encodes an enzyme selected from: glutamate decarboxylase; putrescine aminotransferase; gamma-aminobutyraldehyde dehydrogenase; arginine decarboxylase; agmatinase; ornithine decarboxylase; or a combination thereof. In some embodiments, the glutamate decarboxylase; putrescine aminotransferase; gamma-aminobutyraldehyde dehydrogenase; arginine decarboxylase; agmatinase; ornithine decarboxylase; or a combination thereof, is encoded by a DNA sequence at least 70% similar in DNA sequence to any one of Seq. ID. Nos. 275-304 set forth in Table 3.

(13) In some embodiments, the glutamate decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 4. In some embodiments, the putrescine aminotransferase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 5. In some embodiments, the gamma-aminobutyraldehyde dehydrogenase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 6. In some embodiments, the arginine decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 7. In some embodiments, the agmatinase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 8. In some embodiments, the ornithine decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 9.

(14) In some embodiments, the at least one purified bacterial population consists of bacteria comprising a 16S rDNA sequence having at least 95% similarity to a reference bacterium selected from the group consisting of: *Escherichia coli* MG1655; *Escherichia coli* Nissle 1917; or a combination thereof.

(15) In some embodiments, the bacterial population consists of bacteria capable of producing GABA at a physiologically relevant pH. In some embodiments, the bacterial population consists of bacteria capable of producing GABA at a pH range between about 4.5 and about 7.5. In some embodiments, the bacterial population consists of bacteria capable of producing GABA inside the human gut.

(16) In some embodiments, the composition is in the form of a probiotic, prebiotic, a capsule, a tablet, a caplet, a pill, a troche, a lozenge, a powders, a granule, a medical food, or a combination thereof. In some embodiments, the composition is administered as a fecal transplant.

(17) In some embodiments, the bacteria are capable of producing GABA via expression of any

combination of glutamate decarboxylase, putrescine aminotransferase, gamma-aminobutyraldehyde dehydrogenase, arginine decarboxylase, agmatinase, and/or ornithine decarboxylase.

(18) In some embodiments, the therapeutic composition further comprises a purified bacterial strain that is cytotoxic or cytostatic to a GABA-consuming bacteria. In some embodiments, the GABA-consuming bacteria is *Evtepia gabavorous* or Firmicutes bacterium MGS:114.

(19) In some embodiments, the therapeutic composition further comprises a prebiotic capable of stimulating the growth or GABA-production levels of a GABA-producing bacteria.

(20) In one aspect, the present disclosure provides a method of treating a disease or disorder in a subject in need thereof, the method comprising administering to the subject a therapeutic composition comprising at least one purified bacterial population consisting of bacteria capable of producing GABA in a subject in need thereof.

(21) In one aspect, the present disclosure provides the use of a therapeutic composition comprising at least one purified bacterial population consisting of bacteria capable of producing GABA in the manufacture of a medicament for the treatment of a disease.

(22) In one aspect, the present disclosure provides the use of a therapeutic composition comprising at least one purified bacterial population consisting of bacteria capable of producing GABA for the treatment of a disease.

(23) In some embodiments, the disease or disorder is a mental disease or disorder. In some embodiments, the mental disease or disorder is selected from the group consisting of depression, bipolar disorder, schizophrenia, anxiety, anxiety disorders, addiction, social phobia, treatment-resistant major depressive disorder (TR-MDD), major depressive disorder and its subtypes (melancholic depression, atypical depression, catatonic depression, postpartum depression, and seasonal affective disorder), Neurodegenerative amyloid disorders (Parkinson's, Alzheimer's, and Huntington's diseases) orthostatic tremor, Lafora disease, restless leg syndrome, neuropathic pain, pain disorders, dementia, epilepsy, stiff-person syndrome, premenstrual dysphoric disorder, autism spectrum disorder, sleep disorders, and attention deficit hyperactivity disorder (ADHD), and combinations thereof. In some embodiments, treating a disease or disorder comprises decreasing at least one symptom of the disease or disorder, such as fatigue, insomnia, motor dysfunction, stress, persistent anxiety, persistent sadness, social withdrawal, substance withdrawal, irritability, thoughts of suicide, thoughts of self-harm, restlessness, low sex drive, lack of focus, seizures, memory loss, anger, bouts of emotional reactivity, confusion, pain, and muscle spasms, loss of appetite, altered intestine motility, and combinations thereof.

(24) In some embodiments, the at least one bacterial population consists of a bacteria comprising a 16S rDNA sequence at least about 95% identical to a 16S rDNA sequence selected from one of Seq. ID. Nos. of 1-31 set forth in Table 1. In some embodiments, the at least one purified bacterial population consists of bacteria selected from the group consisting of: *Bacteroides caccae* KLE1911; *Bacteroides clarus* KLE1930; *Bacteroides dorei* KLE1912; *Bacteroides finegoldii* KLE1931; *Bacteroides fragilis* KLE1958; *Bacteroides massiliensis* KLE1932; *Bacteroides ovatus* KLE1770; *Bacteroides stercoris* KLE1933; *Bacteroides thetaiotaomicron* KLE1934; *Bacteroides uniformis* KLE1913; *Bacteroides vulgatus* KLE1910; *Bacteroides xylanisolvens* KLE1935; *Bifidobacterium adolescentis* KLE 1879; *Blautia obeum* KLE1914; *Blautia wexlerae* KLE1916; *Butyricimonas virosa* KLE1938; *Clostridium perfringens* KLE1937; *Clostridium sordellii* KLE1939; *Clostridium* sp. KLE1862; *Clostridium* sp. KLE1918; *Coprobacillus* sp. KLE1779; *Coprococcus* sp. KLE1880; *Dorea longicatena* KLE1917; *Eggerthella lenta* KLE1926; *Eubacterium rectale* KLE1922; *Gordonibacter pamelaee* KLE1915; *Oscillibacter* sp. KLE1928; *Parabacteroides distasonis* KLE2020; *Parabacteroides merdae* KLE1863; *Ruminococcus gnavus* KLE1940; *Turicibacter sanguinis* KLE1941, and combinations thereof.

(25) In some embodiments, the at least one purified bacterial population consists of a bacteria comprising a 16S rDNA sequence at least about 95% identical to a 16S rDNA sequence selected from one of Seq. ID. Nos. of 32-274 set forth in Table 2.

(26) In some embodiments, the at least one bacterial population consists of a bacteria comprising a 16S rDNA sequence at least about 95% identical to any one of Seq ID Nos. 305-2217 set forth in Table 10.

(27) In some embodiments, the at least one purified bacterial population consists of bacteria comprising a DNA sequence which encodes an enzyme selected from: glutamate decarboxylase; putrescine aminotransferase; gamma-aminobutyraldehyde dehydrogenase; arginine decarboxylase; agmatinase; ornithine decarboxylase; or a combination thereof.

(28) In some embodiments, the glutamate decarboxylase; putrescine aminotransferase; gamma-aminobutyraldehyde dehydrogenase; arginine decarboxylase; agmatinase; ornithine decarboxylase; or a combination thereof, is encoded by a DNA sequence at least 70% similar to a DNA sequence selected from one of Seq. ID. Nos. 275-304 set forth in Table 3.

(29) In some embodiments, the glutamate decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 4. In some embodiments, the putrescine aminotransferase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 5. In some embodiments, the gamma-aminobutyraldehyde dehydrogenase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 6. In some embodiments, the arginine decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 7. In some embodiments, the agmatinase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 8. In some embodiments, the ornithine decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 9.

(30) In some embodiments, the bacteria is genetically engineered to produce GABA. In some embodiments, the bacteria is engineered to produce GABA via expression of glutamate decarboxylase; putrescine aminotransferase; gamma-aminobutyraldehyde dehydrogenase; arginine decarboxylase; agmatinase; ornithine decarboxylase; or a combination thereof.

(31) In some embodiments, wherein the glutamate decarboxylase; putrescine aminotransferase; gamma-aminobutyraldehyde dehydrogenase; arginine decarboxylase; agmatinase; ornithine decarboxylase; or a combination thereof, is encoded by a DNA sequence at least 70% similar in DNA sequence selected from one of Seq. ID. Nos. of 275-304 set forth in Table 3.

(32) In some embodiments, the glutamate decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 4. In some embodiments, the putrescine aminotransferase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 5. In some embodiments, the gamma-aminobutyraldehyde dehydrogenase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 6. In some embodiments, the arginine decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 7. In some embodiments, the agmatinase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 8. In some embodiments, the ornithine decarboxylase is encoded by a DNA sequence at least 95% similar in DNA sequence to a gene with a EMBL/GENBANK/DDBJ ID found in Table 9.

(33) In some embodiments, the at least one purified bacterial population consists of bacteria comprising a 16S rDNA sequence having at least 95% similarity to a reference bacterium selected from the group consisting of: *Escherichia coli* MG1655; *Escherichia coli* Nissle 1917; or a combination thereof.

(34) In some embodiments, the bacterial population consists of bacteria capable of producing GABA at a physiologically relevant pH. In some embodiments, the bacterial population consists of bacteria capable of producing GABA at a pH range between about 4.5 and about 7.5. In some

embodiments, bacterial population consists of bacteria capable of producing GABA inside the human gut. In some embodiments, the composition is administered as a fecal transplant. In some embodiments, the composition is administered as a probiotic. In some embodiments, the bacteria are capable of producing GABA via expression any combination of glutamate decarboxylase, putrescine aminotransferase, gamma-aminobutyraldehyde dehydrogenase, arginine decarboxylase, agmatinase, ornithine decarboxylase, and combinations thereof.

(35) In some embodiments, the at least one bacterial strain is cytotoxic or cytostatic to a GABA-consuming bacteria. In some embodiments, the GABA-consuming bacteria is *Evtepia gabavorous* or Firmicutes bacterium MGS:114.

(36) In some embodiments, the method of treating a subject further comprises identifying a subject in need of treatment by determining whether the subject would benefit from an increase in endogenous GABA by measuring an initial amount of GABA in the subject's stool. In some embodiments, the initial amount of GABA in the subject's stool is below about 8 µg per gram of wet or dry stool. In some embodiments, the amount of GABA in the subject's stool is increased relative to the initial amount after administering the therapeutic composition.

(37) In some embodiments, the method of treating a subject further comprises identifying a subject in need of treatment by determining whether the subject would benefit from an increase in endogenous GABA by measuring an initial amount of GABA-producing bacteria in the subject's stool. In some embodiments, the initial amount of GABA-producing bacteria in the subject's stool is less than about 10% of total bacteria as measured by 16S sequence mapping. In some embodiments, at least one GABA-producing bacteria is increased in the subject's stool relative to the initial amount of GABA-producing bacteria in the subject's stool after administering the therapeutic composition.

(38) In some embodiments, the method of treating a subject further comprises identifying a subject in need of treatment by determining whether the subject would benefit from an increase in endogenous GABA by measuring an initial amount of GABA in the subject's blood or serum. In some embodiments, the amount of GABA in the subject's blood or serum is below about 10 µg per liter of blood. In some embodiments, the amount of GABA in the subject's blood or serum is increased relative to the initial amount after administering the therapeutic composition.

(39) In some embodiments, the method of treating a subject further comprises identifying a subject in need of treatment by determining whether the subject would benefit from an increase in endogenous GABA by measuring an amount of GABA in the subject's brain. In some embodiments, the amount of GABA in the subject's brain is below about 1.0 mM/kg. In some embodiments, the amount of GABA in the subject's brain is increased relative to the initial amount after administering the therapeutic composition.

(40) In some embodiments, the method of treating a subject further comprises identifying a subject in need of treatment by determining whether the subject would benefit from an increase in endogenous GABA by measuring an initial amount of expression of GABA-producing enzymes in the subject's stool. In some embodiments, the GABA-producing enzymes are selected from glutamate decarboxylase, putrescine aminotransferase, gamma-aminobutyraldehyde dehydrogenase, arginine decarboxylase, agmatinase, ornithine decarboxylase, and combinations thereof. In some embodiments, the initial amount of enzyme expression is measured by qPCR. In some embodiments, the expression of enzymes is increased relative to the initial amount of enzyme expression after administering the therapeutic composition.

(41) In some embodiments, the method of treating a subject further comprises identifying a subject in need of treatment by determining whether the subject would benefit from an increase in endogenous GABA by measuring an initial amount of GABAergic response in the subject's brain. In some embodiments, the amount of the GABAergic response in the subject's brain is increased relative to the initial amount after administering the therapeutic composition. In some embodiments, the therapeutic composition comprises a prebiotic capable of stimulating the growth

or GABA production of GABA-producing bacteria.

(42) In one aspect, the present disclosure provides a method of culturing a GABA-dependent bacteria, comprising disposing at least one live GABA-dependent bacterial cell on a suitable substrate, and providing a source of GABA.

(43) In some embodiments, the suitable substrate is agar. In some embodiments, providing a source of GABA comprises co-culturing with another bacterial strain, said strain is capable of producing GABA. In some embodiments, GABA is added to the substrate. In some embodiments, the GABA-dependent bacteria is *E. gabavorous*.

(44) In one aspect, the present disclosure provides a method of identifying a bacterial strain or strains capable of producing GABA at a physiologically relevant pH of the human intestinal tract, comprising: (a) dispersing a sample believed to contain GABA-producing bacteria within a substrate, the substrate being at least partially permeable to GABA, (b) contacting the substrate loaded with potential GABA-producing bacteria with a GABA-dependent bacterium; and (c) identifying a GABA-producing bacteria by observing the formation of a colonies of the GABA-dependent bacteria around potential GABA-producing bacteria in the substrate.

(45) In some embodiments, the substrate is being buffered to maintain the pH at a physiologically range found in the human gastrointestinal tract. In some embodiments, the GABA-dependent bacteria is *Egabavorous*. In some embodiments, the pH range is between about 4.5 and about 7.5.

(46) The present disclosure provides compositions and methods for treating mental illness or disease of the central nervous system in a subject and therapeutic compositions for the same. The method can comprise administering to the subject a bacterium and/or multiple bacteria capable of producing endogenous GABA in the subject's gut at a physiologically relevant pH. The present technology can have the advantage of alleviating the symptoms of a mental illness or disease of the central nervous system without the aid of synthetic medications (e.g., antidepressants), which can have unwanted side-effects, or in combination with existing medications. Additionally, the present technology can have the advantage of further improving the digestive health of the subject, such as improving intestinal motility and reducing gastrointestinal pain. Additional features and advantages of the present technology will be apparent to one of skill in the art upon reading the Detailed Description of the Invention, below.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1A is a pair of images (left and top right) that show a co-culture assay to grow and identify previously unculturable bacteria. The inset at the top right shows a close-up of the indicated region of the image at the left.

(2) FIG. 1B shows the growth of *Evtepia gabavorous* KLE1738 in the presence of the helper bacteria *Bacteroides fragilis* KLE1758.

(3) FIG. 2A shows the growth of *Evtepia gabavorous* KLE1738 in the presence of the supernatant from *Bacteroides fragilis*.

(4) FIG. 2B shows a lack of growth of *Evtepia gabavorous* KLE1738 in the presence of the empty vehicle.

(5) FIG. 2C shows the growth of *Evtepia gabavorous* in the presence of the most polar fraction of the supernatant from *Bacteroides fragilis* KLE1758.

(6) FIG. 2D shows a close-view of the lack of growth of *Evtepia gabavorous* KLE1738 in the presence of the empty vehicle.

(7) FIG. 2E shows a close-view of the growth of *Evtepia gabavorous* KLE1738 in the presence of the supernatant from *Bacteroides fragilis* KLE1758.

(8) FIG. 2F shows a close-view of the lack of growth of *Evtepia gabavorous* KLE1738 in the

presence of the empty vehicle (bacterial media).

(9) FIG. 2G shows a table of potential growth factors for *E. gabavorous* KLE1738, with green indicating induction of growth.

(10) FIG. 3 shows a proposed GABA metabolism of *E. gabavorous* KLE1738.

(11) FIG. 4A shows GABA production of *B. fragilis* KLE1758 is observed only at a pH of 5.5 and below, highlighting the importance of acidic pH on GABA production for certain strains or species of bacteria.

(12) FIG. 4B shows an assay for the identification of GABA-producing bacteria, capable of producing GABA at a pH between 4.5 and 7.5, using the GABA-requirement of *E. gabavorous* KLE1738. The medium was heavily buffered to maintain a pH of the desired range and identify strains of bacteria capable of producing GABA at a pH physiologically relevant to the human gut.

(13) FIG. 4C shows a phylogenetic tree of GABA producers, capable of producing GABA at a pH of about 4.5 to about 7.5, identified by the methods herein.

(14) FIG. 5 shows GABA production capabilities of several strains identified using the screen described in Example 7. The ability of organisms to produce GABA at a range of pH conditions is highlighted.

(15) FIG. 6A shows the growth of *E. gabavorous* KLE1738 in the presence of *E. coli* that has been engineered to express glutamate decarboxylase gadA.

(16) FIG. 6B shows the growth of *E. gabavorous* KLE1738 in the presence of *E. coli* that has been engineered to express glutamate decarboxylase gadB.

(17) FIG. 6C shows the growth of *E. gabavorous* KLE1738 in the presence of *B. fragilis* KLE 1758 that is known to produce GABA.

(18) FIG. 6D shows the absence of growth of *E. gabavorous* KLE1738 in the presence of *E. coli* that has been engineered to express glutamate decarboxylase gadC.

(19) FIG. 6E shows the absence of growth of *E. gabavorous* KLE1738 in the presence empty vector.

DETAILED DESCRIPTION OF THE INVENTION

(20) The present disclosure relates to compositions and methods for treating or decreasing a symptom of a disease in a subject. The disease can be a mental illness or a disease of the central nervous system. After identifying a subject with a mental illness or disease of the central nervous system, the method can comprise determining whether the subject would benefit from an increase in endogenous GABA, for instance by measuring the amount of GABA in the subject's stool, blood serum, or other bodily fluids, measuring levels of GABA in different regions of the brain, measuring the GABAergic response in different regions in the brain, measuring activity of GABA producing enzymes in stool, or by measuring the amount of GABA-producing bacteria in the subject's stool. The method can further comprise administering to the subject a GABA-producing bacterium or bacteria that can be capable of producing GABA in the subject's gut, (e.g., at a physiologically relevant pH of the gut).

(21) Some bacteria produce GABA from gamma-aminobutyrate to maintain intracellular pH homeostasis in order to overcome acid stress. As set forth herein, the production of GABA by microbes (e.g., bacteria) in the human gut can impact the health of a subject. For instance, GABA produced by bacteria in the human gut can act as a neurotransmitter to treat a mental illness, a disease of the central nervous system, or improve gastrointestinal health in a subject.

Definitions

(22) As used herein “administer” and “administration” encompasses embodiments in which one person directs another to consume a bacteria or a bacterial composition in a certain manner and/or for a certain purpose, and also situations in which a user uses bacteria or a bacterial composition in a certain manner and/or for a certain purpose independently of or in variance to any instructions received from a second person. Non-limiting examples of embodiments in which one person directs another to consume a bacteria or bacterial composition in a certain manner and/or for a certain

purpose include when a physician prescribes a course of conduct and/or treatment to a patient, when a parent commands a minor user (such as a child) to consume bacteria or a bacterial composition, when a trainer advises a user (such as an athlete) to follow a particular course of conduct and/or treatment, and when a manufacturer, distributor, or marketer recommends conditions of use to an end user, for example through advertisements or labeling on packaging or on other materials provided in association with the sale or marketing of a product.

(23) The term “isolated” encompasses a bacterium or other entity or substance that has been (1) separated from at least some of the components with which it was associated when initially produced (whether in nature, such as human stool, or in an experimental setting, such as a Petri plate consisting of artificial growth medium), and/or (2) produced, prepared, purified, and/or manufactured by the hand of man. Isolated bacteria may be separated from at least about 10%, about 20%, about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90%, or more of the other components with which they were initially associated. In some embodiments, isolated bacteria are more than about 80%, about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more than about 99% pure. As used herein, a substance is “pure” if it is substantially free of other components (such as other bacterial species). The terms “purify,” “purifying” and “purified” refer to a bacterium or other material that has been separated from at least some of the components with which it was associated either when initially produced or generated (e.g., whether in nature or in an experimental setting), or during any time after its initial production, as recognized by those skilled in the art of bacterial cultivation. A bacterium or a bacterial population may be considered purified if it is isolated at or after production, such as from a material or environment containing the bacterium or bacterial population, and a purified bacterium or bacterial population may contain other materials up to about 10%, about 20%, about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90%, or above about 90% and still be considered “isolated.” In some embodiments, purified bacteria and bacterial populations are more than about 80%, about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more than about 99% pure. In the instance of bacterial compositions provided herein, the one or more bacterial types present in the composition can be independently purified from one or more other bacteria produced and/or present in the material or environment containing the bacterial type. Bacterial compositions and the bacterial components thereof are generally purified from residual habitat products.

(24) As used herein, “probiotic” is understood to mean “Live microorganisms which when administered in adequate amounts confer a health benefit on the host”, as currently defined by the World Health Organization.

(25) As used herein, “prebiotic” is understood to mean an ingredient that allows specific changes, both in the composition and/or activity in the gastrointestinal microbiota that may (or may not) confer benefits upon the host.

(26) As used herein, “medical food” is understood to mean “a food which is formulated to be consumed or administered enterally under the supervision of a physician and which is intended for the specific dietary management of a disease or condition for which distinctive nutritional requirements, based on recognized scientific principles, are established by medical evaluation”, as defined by 5(b) of the Orphan Drug Act (21 U.S.C. 360ee (b) (3)).

(27) As used herein “initial amount” is understood to mean the amount of a substance, e.g., GABA in an aliquot of sample prior to administration of GABA-producing bacteria to the subject. Initial amount can be measured in terms of concentration. For instance, an initial amount can be measured in terms of micrograms of substance per milliliter of sample, e.g., micrograms of GABA per milliliter of blood or serum (μg GABA/mL blood or serum). The initial amount of can also be measured, for instance, as the amount of GABA in regions of the brain, such as the prefrontal cortex prior to administration of the GABA-producing bacteria. The amount of GABA can be

represented in terms of millimoles of GABA per kg tissue (mmol GABA per kg of brain tissue). The initial amount can also be measured, for instance, as the amount of GABA in a subject's stool sample prior to administration of GABA-producing bacteria to the subject. The amount of GABA can be represented in terms of micrograms of GABA per gram of stool (μg GABA/g stool). The initial amount can also be the level of expression of GABA producing enzymes in the stool (log change of reads), as measured by qPCR or other appropriate method. Unless otherwise defined herein, stool is weighed when wet or dry, i.e., without active drying, and within one hour of production of the stool. For instance, the stool can be weighed within 45 minutes, 30 minutes, 15 minutes, 10 minutes, or within 5 minutes of production of the stool.

(28) As used herein, a “GABAergic response” means the response of a given organ (e.g., the brain or vagus nerve) to differences in the concentrations of GABA, GABA producing bacteria, or prebiotics to which it is exposed. A GABAergic response can include a change in concentrations of GABA as well as expression levels and/or activity of different GABA.sub.A, GABA.sub.B, and/or GABA.sub.C receptors.

(29) “GABA-producing bacteria” is understood to mean bacteria that can produce measurable quantities of GABA, as detected by LC/MS, ELISA, or other appropriate analytical assays. In some embodiments, GABA-producing bacteria can produce GABA under the physiological conditions in a human, e.g., under the pH, and temperature of the human gut.

(30) “Physiologically relevant pH” of the human intestinal tract is understood to mean a pH range that exists in the body. For instance, a pH range that is physiologically relevant to the human gut can be in the range of about 4.5 to about 7.5.

(31) The term “gut” is understood to refer to the human gastrointestinal tract, also known as the alimentary canal. The gut includes the mouth, pharynx, oesophagus, stomach, small intestine (duodenum, jejunum, ileum), large intestines (cecum and colon) and rectum.

(32) As used herein, “bacteria” or “bacterial strain” is understood to mean a species of bacteria. A “bacterium” is understood as a single bacterial cell of a given species.

(33) The term “treating” with regard to a subject, refers to improving at least one symptom of the subject's disorder. Treating includes curing, improving, or at least partially ameliorating the disorder.

(34) The term “EMBL/GenBank/DDBJ ID” refers to an accession number, which when used as in input for databases such as the European Molecular Biology Laboratory (EMBL), GenBank, or DNA Data Bank of Japan (DDBJ), via their respective internet websites, enables access to information, such as nucleotide sequence of a gene, and the bacterium encoding that sequence in its genome. The EMBL/GenBank/DDBJ ID is used in this application as a convenient means to access sequence information.

(35) Gamma-Aminobutyric Acid (GABA)

(36) The term “GABA” is understood to mean gamma-aminobutyric acid (γ -aminobutyric acid). GABA has the chemical structure:

(37) ##STR00001##

(38) GABA is a major inhibitory neurotransmitter in the mammalian central nervous system. It plays a principal role in reducing neuronal excitability throughout the nervous system. GABA can be a difficult compound to deliver therapeutically due to efflux and half-life limitations. For example, in rodents, the brain efflux rate of GABA was found to be 17 times higher than the influx rate. Additionally, GABA only has a half-life of only 17 minutes in mice. Accordingly, because of the short half-life of GABA in vivo, oral GABA supplementation can be ineffective, as it may require frequent dosing, even with slow release capsules.

(39) The present disclosure provides for delivering a therapeutic composition of one or more bacteria that can produce GABA within the intestinal tract, in order to consistently deliver GABA into systemic circulation (e.g., into the nervous system). The endogenously produced GABA can also activate vagal nerve receptors directly. This can mitigate the inherent half-life of GABA.

(40) In some embodiments, the microbiome can influence GABA levels and the GABAergic response in the brain. For instance, germ-free animals can have substantially reduced luminal and serum levels of GABA. Without wishing to be bound by theory, this suggests the microbiome is important in regulating levels of this important neurotransmitter. As set forth herein, GABAergic modulation by microbiome intervention (e.g., using methods and compositions described herein) can have therapeutic potential.

(41) In some embodiments, GABA can play a role in mental illness or disease of the central nervous system. For instance, in some embodiments, low levels of GABA can be associated with depression, bipolar disorder, schizophrenia, anxiety, anxiety disorders, addiction, social phobia, treatment-resistant major depressive disorder (TR-MDD), major depressive disorder and its subtypes (melancholic depression, atypical depression, catatonic depression, postpartum depression, and seasonal affective disorder), Neurodegenerative amyloid disorders (Parkinson's, Alzheimer's, and Huntington's diseases) orthostatic tremor, Lafora disease, restless leg syndrome, neuropathic pain, pain disorders, dementia, epilepsy, stiff-person syndrome, premenstrual dysphoric disorder, autism spectrum disorder, sleep disorders, and attention deficit hyperactivity disorder (ADHD). As set forth herein, the present disclosure provides increasing the amount of endogenous GABA in a subject can decrease levels of mental illness or disease of the central nervous system in the subject.

(42) In some embodiments, GABA produced by gut bacteria can play a role in mental illness or disease of the central nervous system via the vagus nerve, connecting the intestinal tract to the peripheral and central nervous systems.

(43) In some embodiments, GABA produced by gut bacteria can play a role in mental illness or disease of the central nervous system via affecting circulating levels of systemic GABA in the host, which can influence the peripheral and central nervous systems.

(44) Means of GABA Production by Microbes

(45) Bacteria can produce GABA using a variety of different pathways. Set forth below are exemplary pathways that bacteria and other microbes can use to produce GABA (e.g., in vivo). As set forth below, any of the GABA production pathways described herein can be naturally occurring in a given bacterium. Alternatively, a necessary enzyme or grouping of enzymes can be added to the DNA sequence of a bacteria to enable the bacteria to produce GABA.

(46) Glutamate Pathway

(47) In some embodiments, microbes can produce GABA using the glutamate decarboxylase enzyme (e.g., glutamate decarboxylase EC 4.1.1.15). In some embodiments, glutamate decarboxylase is capable of directly converting glutamate to GABA.

(48) Putrescine to 4-Aminobutanal Pathway

(49) In some embodiments, microbes can produce GABA using the putrescine to 4-aminobutanal pathway. The microbes can then convert 4-aminobutanal to GABA. In some embodiments, putrescine aminotransferase (for instance, putrescine aminotransferase EC 2.6.1.82) can be used to convert putrescine to 4-aminobutanal. The 4-aminobutanal can then be converted in the presence of gamma-aminobutyraldehyde dehydrogenase (e.g., gamma-aminobutyraldehyde dehydrogenase (EC 1.2.1.19)) to GABA.

(50) Arginine to Agmatine to Putrescine Pathway

(51) In some embodiments, microbes can produce GABA using the arginine to agmatine to putrescine Pathway. Once the putrescine is produced, it can be converted as described above (e.g., using the putrescine to 4-aminobutanal pathway) to GABA. In some embodiments, arginine decarboxylase (e.g., arginine decarboxylase (EC 4.1.1.19)) can convert arginine to agmatine. Agmatine can then be converted to putrescine using agmatinase (e.g., agmatinase (EC 3.5.3.11)).

(52) L-Omithine to Putrescine Pathway

(53) In some embodiments, ornithine decarboxylase (e.g., ornithine decarboxylase (EC 4.1.1.17)) can be used to convert ornithine to putrescine. Once the putrescine is produced, it can be converted

as described above (e.g., using the putrescine to 4-aminobutanol pathway) to GABA.

(54) Bacterial Strains

(55) The present disclosure provides bacterial strains (e.g., purified strains) and therapeutic compositions comprising the same for administration to a subject in need thereof. The bacteria can be naturally occurring, or can be engineered (e.g., through strain engineering or selection) to produce GABA. In some embodiments, one strain of GABA-producing bacteria can be administered to a subject. In some embodiments, multiple strains of GABA-producing bacteria can be administered to a subject in need thereof. In some embodiments, the one or more bacteria (e.g., purified bacteria) can act synergistically. For instance, the multiple bacteria can act synergistically to produce high levels of GABA. In some embodiments, the one or more bacteria can also help to reduce the number of GABA-consuming bacteria in a human gut. Accordingly, any one, or any combination of the GABA-producing bacteria taught herein can be administered to a subject in need thereof.

(56) In some embodiments, the bacteria taught herein can produce GABA at physiologically relevant conditions, such as under the conditions of the human gut. In some embodiments, the GABA-producing bacteria taught herein can produce GABA at a pH relevant to the human gut is between about 4.5 and about 7.5. For instance, the pH can be about 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, or any value between about 4.5 and 7.5.

(57) The ability to produce GABA at physiologically relevant pH is important in view of the pH restrictions of GABA production in many bacteria. For example, *E. coli* is unable to produce GABA above a pH of 4.5. Instead, without wishing to be bound by theory, it can use conversion of glutamate to GABA coupled with GABA export, for example, as a means to neutralize the intracellular environment. Accordingly the present disclosure bacteria capable of producing GABA at a pH relevant to the host environment, such as the more neutral large intestines (e.g., between about pH 4.5 and about pH 7.5). In some embodiments, the bacteria taught herein can also produce GABA under other relevant conditions that exist within the human gut. That is, the GABA-producing bacteria taught herein can produce GABA in the absence of oxygen, in a nutrient-competitive and fluctuating environment, and in the absence of light.

(58) Natural Strains

(59) In some embodiments, the GABA-producing bacteria can be identified by having a 16S nucleic acid sequence substantially similar to the 16S sequences of reference bacteria listed in Table 1, with a Seq. ID. No. of 1-31. In some embodiments, the GABA-producing bacteria can have at least 90% 16S sequence similarity to the 16S sequences given in Table 1 (e.g., at least 91% similarity, at least 92% similarity, at least 93% similarity, at least 94% similarity, at least 95% similarity, at least 96% similarity, at least 97% similarity, at least 98% similarity, at least 99% similarity, at least 99.5% similarity, at least 99.9% similarity, or 100% similarity).

(60) In some embodiments, the GABA-producing bacteria can be identified by NCBI Taxon ID of the organism they are most related to by 16S sequence. These bacteria include those belonging to the same NCBI Taxon assignment as *Bacteroides caccae* KLE1911; *Bacteroides clarus* KLE1930; *Bacteroides dorei* KLE1912; *Bacteroides finegoldii* KLE1931; *Bacteroides fragilis* KLE1958; *Bacteroides massiliensis* KLE1932; *Bacteroides ovatus* KLE1770; *Bacteroides stercoris* KLE1933; *Bacteroides thetaiotaomicron* KLE1934; *Bacteroides uniformis* KLE1913; *Bacteroides vulgatus* KLE1910; *Bacteroides xylanisolvens* KLE1935; *Bifidobacterium adolescentis* KLE 1879; *Blautia obeum* KLE1914; *Blautia wexlerae* KLE1916; *Butyrivibrio virosa* KLE1938; *Clostridium perfringens* KLE1937; *Clostridium sordellii* KLE1939; *Clostridium* sp. KLE1862; *Clostridium* sp. KLE1918; *Coprobacillus* sp. KLE1779; *Coprococcus* sp. KLE1880; *Dorea longicatena* KLE1917; *Eggerthella lenta* KLE1926; *Eubacterium rectale* KLE1922; *Gordonibacter pamelaee* KLE1915; *Oscillibacter* sp. KLE1928; *Parabacteroides distasonis* KLE2020; *Parabacteroides merdae* KLE1863; *Ruminococcus gnavus* KLE1940; *Turicibacter sanguinis* KLE1941, and combinations thereof.

(61) TABLE-US-00001 TABLE 1 Strain name of GABA producing bacteria, and their cooresponding Seq. ID No. Strain Seq. ID No. *Bacteroides caccae* KLE1911 1 *Bacteroides clarus* KLE1930 2 *Bacteroides dorei* KLE1912 3 *Bacteroides finegoldii* KLE1931 4 *Bacteroides fragilis* KLE1958 5 *Bacteroides massiliensis* KLE1932 6 *Bacteroides ovatus* KLE1770 7 *Bacteroides stercoris* KLE1933 8 *Bacteroides thetaiotaomicron* KLE1934 9 *Bacteroides uniformis* KLE1913 10 *Bacteroides vulgatus* KLE1910 11 *Bacteroides xylanisolvens* KLE1935 12 *Bifidobacterium adolescentis* KLE1879 13 *Blautia obeum* KLE1914 14 *Blautia wexlerae* KLE1916 15 *Butyricimonas virosa* KLE1938 16 *Clostridium perfringens* KLE1937 17 *Clostridium sordellii* KLE1939 18 *Clostridium* sp. KLE1862 19 *Clostridium* sp. KLE1918 20 *Coprobacillus* sp. KLE1779 21 *Coprococcus* sp. KLE1880 22 *Dorea longicatena* KLE1917 23 *Eggerthella lenta* KLE1926 24 *Eubacterium rectale* KLE1922 25 *Gordonibacter pamelaee* KLE1915 26 *Oscillibacter* sp. KLE1928 27 *Parabacteroides distasonis* KLE2020 28 *Parabacteroides merdae* KLE1863 29 *Ruminococcus gnavus* KLE1940 30 *Turicibacter sanguinis* KLE1941 31

(62) Also disclosed herein are bacteria that are predicted to be capable of producing GABA (e.g., under physiologically relevant conditions and/or in the human gut). Bacteria are identified as being candidate GABA producing bacteria if they have encoded in their genome enzymes involved in GABA biosynthesis. In some embodiments, the bacteria that are predicted to be capable of producing GABA can be identified by having a 16S nucleic acid sequence substantially similar to the 16S sequences of reference bacteria listed in Table 2, with a Seq. ID. No. of 32-274. In some embodiments, the predicted GABA-producing bacteria can have at least 90% 16S sequence similarity to the 16S sequences given in Table 2, with a Seq. ID. No. of 32-274 (e.g., at least 91% similarity, at least 92% similarity, at least 93% similarity, at least 94% similarity, at least 95% similarity, at least 96% similarity, at least 97% similarity, at least 98% similarity, at least 99% similarity, at least 99.5% similarity, at least 99.9% similarity, or 100% similarity).

(63) TABLE-US-00002 TABLE 2 Predicted GABA producing bacteria and their corresponding Seq. ID No.. Species Seq. ID No. *Abiotrophia defective* 32 *Acetobacter okinawensis* 33 *Achromobacter arsenitoxydans* 34 *Achromobacter xylosoxidans* 35 *Acidovorax* sp. 36 *Acidovorax* sp. 37 *Acidovorax* sp. 38 *Acidovorax* sp. 39 *Acidovorax* sp. 40 *Actinoplanes friuliensis* 41 *Aeromonas enteropelogenes* 42 *Aeromonas hydrophila* 43 *Afipia birgiae* 44 *Afipia clevelandensis* 45 *Afipia* sp. 46 *Agrobacterium albertimagni* 47 *Agrobacterium* sp. 48 *Agrobacterium tumefaciens* 49 *Akkermansia muciniphila* 50 *Alcaligenes faecalis* 51 *Alicycophilus denitrificans* 52 *Alistipes finegoldii* 53 *Alistipes indistinctus* 54 *Alistipes onderdonkii* 55 *Alistipes putredinis* 56 *Alistipes shahii* 57 *Aquamicrobium defluvii* 58 *Arenimonas donghaensis* 59 *Arthrobacter* sp. 60 *Azospirillum* sp. 61 *Bacillus bataviensis* 62 *Bacillus cereus* 63 *Bacillus cereus* 64 *Bacillus endophyticus* 65 *Bacillus weihenstephanensis* 66 *Bacteroidaceae bacterium* 67 *Bacteroides acidifaciens* 68 *Bacteroides caccae* 69 *Bacteroides cellulosilyticus* 70 *Bacteroides dorei* 71 *Bacteroides eggerthii* 72 *Bacteroides finegoldii* 73 *Bacteroides fragilis* 74 *Bacteroides gallinarum* 75 *Bacteroides intestinalis* 76 *Bacteroides massiliensis* 77 *Bacteroides oleiciplenus* 78 *Bacteroides ovatus* 79 *Bacteroides rodentium* 80 *Bacteroides salyersiae* 81 *Bacteroides sartorii* 82 *Bacteroides* sp. 83 *Bacteroides* sp. 84 *Bacteroides* sp. 85 *Bacteroides* sp. 86 *Bacteroides stercoris* 87 *Bacteroides thetaiotaomicron* 88 *Bacteroides uniformis* 89 *Bacteroides vulgatus* 90 *Bacteroides xylanisolvens* 91 *Barnesiella intestinihominis* 92 *Bhargavaea cecembensis* 93 *Bifidobacterium adolescentis* 94 *Bifidobacterium angulatum* 95 *Bifidobacterium dentium* 96 *Bifidobacterium ruminantium* 97 *Blastococcus* sp. 98 *Bordetella bronchiseptica* 99 *Bordetella trematum* 100 *Bosea* sp. 101 *Bradyrhizobium* sp. 102 *Brevibacillus borstelensis* 103 *Brevundimonas diminuta* 104 *Brevundimonas naejangsanensis* 105 *Brucella abortus* 106 *Brucella melitensis* 107 *Brucella neotomae* 108 *Burkholderia mallei* 109 *Burkholderia multivorans* 110 *Carnobacterium gallinarum* 111 *Caulobacter crescentus* 112 *Caulobacter* sp. 113 *Cellulomonas flavigena* 114 *Cellulomonas* sp. 115 *Cellulosimicrobium cellulans* 116 *Cetobacterium somerae* 117 *Citrobacter amalonaticus* 118 *Cloacibacillus evryensis* 119 *Clostridium acetobutylicum* 120 *Clostridium perfringens* 121

Comamonas granuli 122 *Corynebacterium variabile* 123 *Cupriavidus basilensis* 124 *Cupriavidus*
sp. 125 *Dechloromonas agitata* 126 *Deinococcus geothermalis* 127 *Delftia tsuruhatensis* 128
Desulfovibrio desulfuricans 129 *Desulfovibrio sp.* 130 *Devosia riboflavina* 131 *Eggerthella sp.* 132
Ensifer adhaerens 133 *Enterococcus casseliflavus* 134 *Enterococcus flavescens* 135 *Escherichia*
coli 136 *Eubacterium limosum* 137 *Eubacterium nodatum* 138 *Eubacterium saphenum* 139
Fusobacterium periodonticum 140 *Gordonia sputi* 141 *Gordonia terrae* 142 *Gordonibacter*
pamelaee 143 *Halomonas stevensii* 144 *Halomonas titanicae* 145 *Hoeflea sp.* 146
Intrasporangium calvum 147 *Janibacter hoylei* 148 *Kaistia granuli* 149 *Kineococcus radiotolerans*
150 *Lactobacillus coleohominis* 151 *Lactobacillus plantarum* 152 *Lactobacillus reuteri* 153
Lactococcus garvieae 154 *Lactococcus lactis* 155 *Lautropia mirabilis* 156 *Leucobacter salsicium*
157 *Luteimonas huabeiensis* 158 *Magnetospirillum magnetotacticum* 159 *Marinobacter lipolyticus*
160 *Marmoricola sp.* 161 *Megasphaera micronuciformis* 162 *Megasphaera sp.* 163 *Mesorhizobium*
sp. 164 *Methanobrevibacter arboriphilus* 165 *Methylobacterium radiotolerans* 166
Methylobacterium sp. 167 *Microbacterium sp.* 168 *Micromonospora aurantiaca* 169
Mogibacterium Mogibacterium 170 *Morganella morganii* 171 *Mycobacterium smegmatis* 172
Mycobacterium sp. 173 *Mycobacterium sp.* 174 *Mycobacterium vanbaalenii* 175 *Neisseria sicca*
176 *Neorhizobium galegae* 177 *Nocardia rhamnosiphila* 178 *Nocardiopsis alkaliphila* 179
Nocardiopsis ganjiahuensis 180 *Nocardiopsis synnemataformans* 181 *Nocardiopsis valliformis* 182
Novosphingobium nitrogenifigens 183 *Ochrobactrum intermedium* 184 *Odoribacter laneus* 185
Odoribacter splanchnicus 186 *Oerskovia turbata* 187 *Pannonibacter phragmitetus* 188 *Pantoea*
vagans 189 *Parabacteroides distasonis* 190 *Parabacteroides goldsteinii* 191 *Parabacteroides*
johnsonii 192 *Parabacteroides merdae* 193 *Parabacteroides sp.* 194 *Parabacteroides sp.* 195
Paracoccus denitrificans 196 *Paracoccus sp.* 197 *Paracoccus yeei* 198 *Parvimonas Parvimonas*
199 *Pectobacterium carotovorum* 200 *Phyllobacterium sp.* 201 *Polaromonas sp.* 202
Porphyromonas bennonis 203 *Proteus mirabilis* 204 *Providencia alcalifaciens* 205 *Providencia*
burhodogranariae 206 *Providencia rettgeri* 207 *Pseudacidovorax intermedium* 208
Pseudoalteromonas sp. 209 *Pseudochrobactrum sp.* 210 *Pseudomonas aeruginosa* 211
Pseudomonas alcaligenes 212 *Pseudomonas chloritidismutans* 213 *Pseudomonas chlororaphis* 214
Pseudomonas japonica 215 *Pseudomonas knackmussii* 216 *Pseudomonas mendocina* 217
Pseudomonas monteilii 218 *Pseudomonas oleovorans* 219 *Pseudomonas putida* 220 *Pseudomonas*
savastanoi 221 *Pseudomonas sp.* 222 *Pseudomonas sp.* 223 *Pseudomonas sp.* 224 *Pseudomonas*
sp. 225 *Pseudomonas sp.* 226 *Pseudomonas sp.* 227 *Pseudomonas sp.* 228 *Pseudomonas sp.* 229
Pseudomonas sp. 230 *Pseudomonas stutzeri* 231 *Pseudomonas synxantha* 232 *Pseudomonas*
syringae 233 *Pseudonocardia sp.* 234 *Ralstonia solanacearum* 235 *Raoultella planticola* 236
Rhizobium leguminosarum 237 *Rhizobium sp.* 238 *Rhodococcus defluvii* 239 *Rhodococcus*
pyridinivorans 240 *Rikenella microfus* 241 *Robinsoniella sp.* 242 *Roseomonas cervicalis* 243
Roseomonas sp. 244 *Salmonella enterica* 245 *Sanguibacter keddiei* 246 *Shewanella baltica* 247
Shewanella sp. 248 *Shinella zoogloeoides* 249 *Sphingopyxis alaskensis* 250 *Starkeya novella* 251
Stenotrophomonas maltophilia 252 *Stenotrophomonas rhizophila* 253 *Streptococcus thermophilus*
254 *Streptomyces atroolivaceus* 255 *Streptomyces coelicollavus* 256 *Streptomyces olindensis* 257
Streptomyces rimosus 258 *Streptomyces roseovorticillatus* 259 *Streptomyces sp.* 260 *Streptomyces*
sp. 261 *Streptomyces sp.* 262 *Streptomyces sp.* 263 *Streptomyces sp.* 264 *Streptomyces sp.* 265
Streptomyces sp. 266 *Streptomyces toyocaensis* 267 *Streptomyces turgidiscabies* 268 *Synergistes*
sp. 269 *Tannerella sp.* 270 *Thauera terpenica* 271 *Variovorax paradoxus* 272 *Variovorax sp.* 273
Xanthomonas axonopodis 274

Engineered Strains

- (64) In some embodiments, bacteria can be engineered to produce GABA (e.g., under the conditions of the human gut). The bacteria can be engineered using techniques of molecular biology, or can be evolved using the process of selection to produce GABA in the human gut.
- (65) As set forth above, GABA can be produced by multiple pathways within a microbial cell. For

example, GABA can be produced by the glutamate pathway, the putrescine to 4-aminobutanol pathway, the arginine to agmatine to putrescine pathway, the L-ornithine to putrescine pathway, or by a combination of pathways. In some embodiments, bacteria can be engineered to contain one or more enzymes in any one of the above pathways that can enable the bacteria to produce GABA or a necessary precursor to GABA.

(66) A variety of different host bacteria can be engineered to produce GABA. For instance, in some embodiments, *Escherichia coli* Nissle 1917, can be genetically modified or selected through evolution to produce GABA. In some embodiments, the bacteria (e.g., *Escherichia coli* Nissle 1917) can be modified to express or overexpress glutamate decarboxylase A or glutamate decarboxylase B. The bacteria can also be made to produce GABA by one or more of the other pathways described herein.

(67) Accordingly, in some embodiments, an engineered GABA-producing strain can be identified as having a specific enzyme encoded in its genomes. For example, the enzyme can be glutamate decarboxylase (EC 4.1.1.15); putrescine aminotransferase (EC 2.6.1.82); gamma-aminobutyraldehyde dehydrogenase (EC 1.2.1.19); arginine decarboxylase (EC 4.1.1.19); agmatinase (EC 3.5.3.11); ornithine decarboxylase (EC 4.1.1.17); or a combination thereof. In some embodiments, the GABA-producing strain can be engineered to contain an enzyme that has at least 50% similarity with with the representative sequences listed in Table 3 (e.g., at least 60% similarity, at least 70% similarity, at least 80% similarity, at least 90% similarity, at least 91% similarity, at least 92% similarity, at least 93% similarity, at least 94% similarity, at least 95% similarity, at least 96% similarity, at least 97% similarity, at least 98% similarity, at least 99% similarity, at least 99.5% similarity, at least 99.9% similarity, or 100% similarity). The enzyme classes, as identified by their Enzyme Commission (EC) numbers, are listed in Table 3.

(68) TABLE-US-00003 TABLE 3 Enzymes involved in GABA-Production Pathways, and Seq. ID No. for representative sequences for each enzyme class Enzyme Seq ID No. Glutamate decarboxylase (EC 4.1.1.15) 275-279 Putrescine aminotransferase (EC 2.6.1.82) 280-284 Gamma-aminobutyraldehyde dehydrogenase (EC 1.2.1.19) 285-289 Arginine decarboxylase (EC 4.1.1.19) 290-294 Agmatinase (EC 3.5.3.11) 295-299 Ornithine decarboxylase (EC 4.1.1.17) 300-304

(69) Representative examples of glutamate decarboxylase (EC 4.1.1.15) are given below in Table 4 and identified by their EMBL/GENBANK/DDBJ ID numbers. Any of the bacteria given in Table 10 can be engineered with any version of the glutamate decarboxylase set forth in Table 3 or Table 4. For instance, the bacteria can be engineered with a version of the glutamate decarboxylase enzyme that has at least 50% nucleotide similarity with any of the versions of glutamate decarboxylase given in Table 4 (e.g., at least nucleotide 60% similarity, at least 70% nucleotide similarity, at least 80% nucleotide similarity, at least 90% nucleotide similarity, at least 91% nucleotide similarity, at least 92% nucleotide similarity, at least 93% nucleotide similarity, at least 94% nucleotide similarity, at least 95% nucleotide similarity, at least 96% nucleotide similarity, at least 97% nucleotide similarity, at least 98% nucleotide similarity, at least 99% nucleotide similarity, at least 99.5% nucleotide similarity, at least 99.9% nucleotide similarity, or 100% nucleotide similarity).

(70) TABLE-US-00004 TABLE 4 Representative Examples of Glutamate Decarboxylase by EMBL/GENBANK/DDBJ ID Number AAA23833.1 ABF03896.1 ABV88270.1 ADI09031.1 AAA23834.1 ABF05994.1 ABV98199.1 ADJ50560.1 AAB18493.1 ABF08645.1 ABW16238.1 ADM42706.1 AAC46188.1 ABG57877.1 ABY23810.1 ADN37174.1 AAC74566.1 ABG82826.1 ABY39350.1 ADO35975.1 AAC76542.1 ABG85385.1 ABZ77591.1 ADP83065.1 AAG22560.1 ABG97793.1 ABZ87126.1 ADR26899.1 AAG22562.1 AB147454.1 ACA46668.1 ADR28903.1 AAG56275.1 ABJ00921.1 ACA53668.1 ADT67741.1 AAG58658.1 ABJ02999.1 ACA88989.1 ADT75096.1 AAK05388.1 ABJ63253.1 ACB15780.1 ADT77123.1 AAK17187.1 ABJ64910.1 ACB15938.1 ADU00909.1 AAK47878.1 ABK67541.1 ACC39574.1 ADU26301.1 AAL54152.1 ABK75920.1 ACD04211.1 ADW04875.1 AAL54153.1 ABK90558.1 ACD07724.1 ADY32347.1

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ADY59978.1 AAN54823.2 ABM09145.1 ACJ29997.1 ADZ09662.1 AAN80380.1 ABM12315.1
ACL47965.1 ADZ09880.1 AAN82764.1 ABM78546.1 ACN16811.1 AEA24253.1 AAO77677.1
ABP47401.1 ACO85906.1 AEA24271.1 AAP17196.1 ABP56802.1 ACR68120.1 AEB72391.1
AAP19142.1 ABQ62050.1 ACS81597.1 AEE27076.1 AAS06807.1 ABQ75258.1 ACU49468.1
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ACU77475.1 AEF37245.1 AAW85387.1 ABQ85039.1 ACV54143.1 AEF40931.1 AAW85559.1
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AIS31569.1 AJI53611.1 AFJ28998.1 AGV73208.1 AIS60683.1 AJI57942.1 AFJ31191.1
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EJ258218.1 ELC41591.1 EMT34264.1 END92583.1 EJ261626.1 ELC88436.1 EMU58101.1
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EMV21034.1 ENH06612.1 EKV49243.1 ELG85654.1 EMV21148.1 ENH09530.1 EKV49295.1
ELG89467.1 EMW92532.1 ENT93039.1 EKV49297.1 ELP68892.1 EMW92626.1 ENZ87560.1
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EOA59672.1 EKV53661.1 ELS31036.1 EMX17914.1 EOD68466.1 EKJ88854.1 ELS48492.1

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EKN18388.1 ELW26442.1 EMX40966.1 EOH76700.1 EKN22714.1 ELW30245.1 EMX45867.1
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(71) Representative examples of putrescine aminotransferase (EC 2.6.1.82) are given below in Table 5 and identified by their EMBL/GenBank/DDBJ ID numbers. Any of the bacteria given in Table 10 can be engineered with any version of the putrescine aminotransferase (EC 2.6.1.82) set forth in Table 3 and Table 5. For instance, the bacteria can be engineered with a version of the putrescine aminotransferase enzyme that has at least 50% nucleotide similarity with any of the versions of putrescine aminotransferase given in Table 5 (e.g., at least nucleotide 60% similarity, at least 70% nucleotide similarity, at least 80% nucleotide similarity, at least 90% nucleotide similarity, at least 91% nucleotide similarity, at least 92% nucleotide similarity, at least 93% nucleotide similarity, at least 94% nucleotide similarity, at least 95% nucleotide similarity, at least 96% nucleotide similarity, at least 97% nucleotide similarity, at least 98% nucleotide similarity, at least 99% nucleotide similarity, at least 99.5% nucleotide similarity, at least 99.9% nucleotide similarity, or 100% nucleotide similarity).

(72) TABLE-US-00005 TABLE 5 Representative Examples of Putrescine Aminotransferase by EMBL/GenBank/DDBJ ID Number AAA57874.1 ACB17060.1 AEJ58471.1 AID92376.1 AAA89152.1 ACF64513.1 AEN66465.1 AII63595.1 AAC36832.1 ACF70070.1 AEQ14325.1 AIP96878.1 AAC45301.1 ACF88918.1 AEW04404.1 AIR04884.1 AAC74384.1 ACH48710.1 AEW63332.1 AIR60611.1 AAC75709.1 ACH77510.1 AEW75317.1 AIR68505.1 AAC76108.3 ACI09453.1 AEX02383.1 AIS03494.1 AAFI0980.1 ACI18235.1 AFC33584.1 AIS53235.1 AAG03522.1 ACI36066.1 AFG42020.1 AIT00514.1 AAG03688.1 ACI77901.1 AFH65907.1 AIU72158.1 AAG58206.1 ACI77902.1 AFI89823.1 AIW78685.1 AAKI5486.1 ACI77903.1 AFJ30752.1 AIW86257.1 AAL22091.1 ACI77904.1 AFJ45596.1 AIX66695.1 AAN44587.2 ACI77905.1 AFJ57021.1 AIZ84136.1 AAN82273.1 ACN47379.1 AFM41617.1 AJA28077.1 AAO70680.1 ACO02631.1 AFM61662.1 AJB64485.1 AAP18399.1 ACO80985.1 AFN34266.1 AJC52685.1 AAV78920.1 ACR65342.1 AFR02837.1 AJC61854.1 AAX67071.1 ACS85106.1 AFS72647.1 AJC65876.1 AAZ89792.1 ACT07527.1 AFT85698.1 AJE19840.1 ABA89323.1 ACT12462.1 AFU12013.1 AJE57644.1 ABB67447.1 ACT27738.1 AGB76663.1 AJE82594.1 ABE08957.1 ACX87602.1 AGF56701.1 AJF57894.1 ABF05173.1 ACY90298.1 AGH86154.1 AJF75010.1 ABG71145.1 ACZ77451.1 AGM28805.1 AJH17424.1 ABJ02583.1 ADA75444.1 AGN85841.1 AJK49269.1 ABJ15261.1 ADD58285.1 AGQ74484.1 AJ086687.1 ABO88369.1 ADF63987.1 AGR60480.1 AJQ98395.1 ABP62186.1 ADM97796.1 AHA09869.1 AJY76773.1 ABR78896.1 ADO46913.1 AHA67171.1 AJZ91824.1 ABU78707.1 ADO70560.1 AHA67172.1 AKE62036.1 ABV07484.1 ADR28470.1 AHB72039.1 AKE93437.1 ABV15541.1 ADT76707.1 AHE60723.1 AKH08854.1 ABV20592.1 ADX19008.1 AHE69148.1 AKH23767.1 ABX24187.1 AEA71912.1 AHF76631.1 AKK34697.1 ABX64022.1 AEE58362.1 AHJ76496.2 AKK40619.1 ABX69344.1 AEE97629.1 AHM77480.1 AKK49917.1 ABX75605.1 AEG95705.1 AHM83064.1 AKL15453.1 ABY42524.1 AEI06121.1 AHN81979.1 AKL34316.1 ACA76303.1 AEI45944.1 AHY10460.1 AKL96946.1 ACB04157.1 AEJ41147.1 AHY72510.1 AKM18742.1 AKM36624.1 BAG78878.1 CCF65898.1 CEG33484.1 AKP48388.1 BAI27353.1 CCI78933.1 CEH27897.1 AKP75846.1 BAI32529.1 CCJ45694.1 CEH28958.1 AKZ54414.1 BAI37675.1 CCJ72687.1 CEJ63776.1 ALA75381.1 BAK33480.1 CCJ77939.1 CEK06962.1 ALB43935.1 BAM00748.1

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(73) Representative examples of gamma-aminobutyraldehyde dehydrogenase (EC 1.2.1.19) are given below in Table 6 and identified by their EMBL/GENBANK/DDBJ ID numbers. Any of the bacteria given in Table 10 can be engineered with any version of the gamma-aminobutyraldehyde dehydrogenase (EC 1.2.1.19) set forth in Table 3 and Table 6. For instance, the bacteria can be engineered with a version of the gamma-aminobutyraldehyde dehydrogenase enzyme that has at least 50% nucleotide similarity with any of the versions of gamma-aminobutyraldehyde dehydrogenase given in Table 6 (e.g., at least nucleotide 60% similarity, at least 70% nucleotide similarity, at least 80% nucleotide similarity, at least 90% nucleotide similarity, at least 91% nucleotide similarity, at least 92% nucleotide similarity, at least 93% nucleotide similarity, at least 94% nucleotide similarity, at least 95% nucleotide similarity, at least 96% nucleotide similarity, at least 97% nucleotide similarity, at least 98% nucleotide similarity, at least 99% nucleotide similarity, at least 99.5% nucleotide similarity, at least 99.9% nucleotide similarity, or 100% nucleotide similarity).

(74) TABLE-US-00006 TABLE 6 Representative Examples of Gamma-Aminobutyraldehyde Dehydrogenase by EMBL/GENBANK/DDBJ ID Number AAA23428.1 AAX65500.1 ABE07141.1 ABO56067.1 AAC74382.1 AAY35300.1 ABE33447.1 ABP60799.1 AAC74526.1 AAY91615.1 ABE35009.1 ABP83011.1 AAG56331.1 AAZ88386.1 ABE58543.1 ABQ62162.1 AAL20515.1 ABA73866.1 ABF03934.1 ABQ79255.1 AAN43345.1 ABA76670.1 ABF07563.1 ABQ81306.1 AAN54343.1 ABA77271.1 ABF08506.1 ABR77355.1 AAN68409.1 ABA81134.1 ABF77276.1 ABR80823.1 AAN80331.1 ABB06719.1 ABG69453.1 ABU77182.1 AAO69141.1 ABB09933.1 ABI88593.1 ABV05856.1 AAP17225.1 ABB61850.1 ABI91617.1 ABV12611.1 AAV77220.1 ABC35115.1 ABJ00886.1 ABV19407.1 ABV19711.1 ADF61409.1 AFG40645.1 AGO56529.1 ABV42738.1 ADF61777.1 AFI68614.1 AGO56534.1 ABV94754.1 ADI12862.1 AFI89822.1 AGP45636.1 ABX21278.1 ADJ47943.1 AFJ28854.1 AGP48796.1 ABX67082.1 ADM97795.1 AFJ28952.1 AGP52381.1 ACA77853.1 ADN75355.1 AFJ47105.1 AGP54692.1 ACB02660.1 ADO48635.1 AFK56420.1 AGP54697.1 ACB16069.1 ADP16410.1 AFK67163.1 AGP60625.1 ACB17411.1 ADR26852.1 AFL53103.1 AGQ73200.1 ACD08717.1 ADT74882.1 AFM59735.1 AGR58814.1 ACF65188.1 ADT75049.1 AFM60144.1 AGS23468.1 ACH48735.1 ADV53576.1 AFN32345.1 AGS72121.1 ACI07089.1 ADX17297.1 AFO48697.1 AGZ34904.1 ACI11169.1 AEC17673.1 AFO86070.1 AGZ34909.1 ACI11654.1 AEE56233.1 AFO92872.1 AGZ36730.1 ACI38546.1 AEE56444.1 AFR02836.1 AHA65135.1 ACI83776.1 AEG95612.1 AFR27692.1 AHA65444.1 ACI83777.1 AEG98095.1 AFR30100.1 AHA65445.1 ACI83778.1 AEG98933.1 AFR31023.1 AHB59062.1 ACI83779.1 AEH78268.1 AFS74542.1 AHB70534.1 ACI83780.1 AEH79912.1 AFS74651.1 AHB70670.1 ACI84176.1 AEI77057.1 AFT90302.1

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(75) Representative examples of arginine decarboxylase (EC 4.1.1.19) are given below in Table 7 and identified by their EMBL/GENBANK/DDBJ ID numbers. Any of the bacteria given in Table 10 can be engineered with any version of the arginine decarboxylase (EC 4.1.1.19) set forth in Table 3 and Table 7. For instance, the bacteria can be engineered with a version of the arginine decarboxylase enzyme that has at least 50% nucleotide similarity with any of the versions of arginine decarboxylase given in Table 7 (e.g., at least nucleotide 60% similarity, at least 70% nucleotide similarity, at least 80% nucleotide similarity, at least 90% nucleotide similarity, at least 91% nucleotide similarity, at least 92% nucleotide similarity, at least 93% nucleotide similarity, at least 94% nucleotide similarity, at least 95% nucleotide similarity, at least 96% nucleotide similarity, at least 97% nucleotide similarity, at least 98% nucleotide similarity, at least 99% nucleotide similarity, at least 99.5% nucleotide similarity, at least 99.9% nucleotide similarity, or 100% nucleotide similarity).

(76) TABLE-US-00007 TABLE 7 Representative Examples of Arginine Decarboxylase by EMBL/GENBANK/DDBJ ID Number AAA23481.1 AAG08224.1 AAN83546.1 AAQ61700.1 AAA24646.1 AAG58069.1 AAO10730.2 AAR35910.1 AAA62686.1 AAK03466.1 AAO28012.1 AAS63667.1 AAA69105.1 AAK65018.1 AAO58271.1 AAS81619.1 AAA97017.1 AAK65019.2 AAO70551.1 AAS94900.1 AAC24937.1 AAL21961.1 AAO71667.1 AAT33294.1 AAC75975.1 AAL23120.1 AAO78500.1 AAT56175.1 AAC77078.2 AAM38760.1 AAP07133.1 AAU16546.1 AAD06541.1 AAM43099.1 AAP10882.1 AAU26114.1 AAD07486.1 AAM86863.1 AAP18235.1 AAU91686.1 AAF09826.1 AAN44412.1 AAP27896.1 AAV78788.1 AAF40905.1 AAN54922.1 AAP77830.1 AAV81611.1 AAF82957.1 AAN66194.1 AAP99095.1 AAV82053.1 AAF96713.1 AAN81972.1 AAQ60544.1 AAV93917.1 AAW73474.1 ABE57906.1 ABM69334.1 ABW26679.1 AAW87910.1 ABF05062.1 ABM71261.1 ABX07146.1 AAW90125.1 ABF09627.1 ABM74623.1 ABX07983.1 AAX66932.1 ABF46066.1 ABM79561.1 ABX24332.1 AAY39411.1 ABF85039.1 ABN07206.1 ABX48976.1 AAY50992.1 ABF88624.1 ABN61275.1 ABX69181.1 AAY96086.1 ABG12304.1 ABN69317.1 ABX73818.1 AAZ33811.1 ABG12881.1 ABO16671.1 ABY34333.1 AAZ58866.1 ABG19102.1 ABO23486.1 ABY96523.1 AAZ89679.1 ABG19453.1 ABO48608.1 ABZ76368.1 ABA21777.1 ABG42373.1 ABO50340.1 ACA11160.1 ABA23030.1 ABG50504.1 ABO90758.1 ACA16086.1 ABA48961.1 ABG50611.1 ABP62006.1 ACA35608.1 ABA57077.1 ABG58847.1 ABP75372.1 ACA67146.1 ABA72371.1 ABI38635.1 ABP78675.1 ACA75078.1 ABA89337.1 ABI42627.1 ABP83495.1 ACA86341.1 ABB10893.1 ABI46161.1 ABQ05833.1 ACA99796.1 ABB26344.1 ABI55429.1 ABQ19043.1 ACB15562.1 ABB27130.1 ABI60410.1 ABQ25189.1 ACB16864.1 ABB31146.1 ABI68665.1 ABQ76774.1 ACB59509.1 ABB35370.1 ABI71549.1 ABQ91208.1 ACB76743.1 ABB36243.1 ABJ02367.1 ABR39026.1 ACB91563.1 ABB37372.1 ABJ14223.1 ABR71752.1 ACC79266.1 ABB44410.1 ABJ88595.1 ABR78771.1 ACC84558.1 ABB49107.1 ABKI5308.1 ABR85787.1 ACD05477.1 ABB50154.1 ABK38605.1 ABR88619.1 ACD08093.1 ABB56739.1 ABK43242.1 ABS07904.1 ACD08437.1 ABB57067.1 ABK47857.1 ABS26167.1 ACD22470.1 ABB63148.1 ABK75092.1 ABS44765.1 ACD56878.1 ABB67557.1 ABK82178.1 ABS51948.1 ACD84161.1 ABB74343.1 ABK86821.1 ABU58554.1 ACD95391.1 ABC18380.1 ABL00211.1 ABU74620.1 ACE82746.1 ABC20552.1 ABL82864.1 ABU75700.1 ACF30404.1 ABC33051.1 ABL89457.1 ABV15373.1 ACF53467.1 ABC81667.1 ABM00169.1 ABV17227.1 ACF61369.1 ABC98358.1 ABM02703.1 ABV17863.1 ACF63192.1 ABD03703.1 ABM13722.1 ABV37216.1 ACF69056.1 ABD80860.1 ABM17621.1 ABV43068.1 ACF90655.1 ABE08775.1 ABM25203.1 ABV49674.1 ACG73211.1 ABE10119.1 ABM29533.1 ABV67224.1 ACH39919.1 ABE55560.1 ABM63175.1 ABV87800.1 ACH51462.1 ACH52106.1 ACO76955.1 ADE16426.1 ADP85550.1 ACH64501.1 ACP07734.1 ADE54552.1 ADP96036.1 ACH76229.1 ACQ68147.1 ADE68385.1 ADQ16388.1 ACH83424.1 ACQ92008.1 ADE82828.1 ADQ41297.1 ACI09897.1 ACQ95950.1 ADF38194.1 ADQ80979.1 ACI17330.1 ACR12004.2 ADF54509.1 ADR18999.1 ACI27705.1 ACR67553.1 ADF63801.1 ADR23218.1 ACI37609.1

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OCS95943.1 SBZ90867.1

(77) Representative examples of agmatinase (EC 3.5.3.11) are given below in Table 8 and identified by their EMBL/GENBANK/DDBJ ID numbers. Any of the bacteria given in Table 10 can be engineered with any version of the agmatinase (EC 3.5.3.11) set forth in Table 3 and Table 8. For instance, the bacteria can be engineered with a version of the agmatinase enzyme that has at least 50% nucleotide similarity with any of the versions of agmatinase given in Table 8 (e.g., at least nucleotide 60% similarity, at least 70% nucleotide similarity, at least 80% nucleotide similarity, at least 90% nucleotide similarity, at least 91% nucleotide similarity, at least 92% nucleotide similarity, at least 93% nucleotide similarity, at least 94% nucleotide similarity, at least 95% nucleotide similarity, at least 96% nucleotide similarity, at least 97% nucleotide similarity, at least 98% nucleotide similarity, at least 99% nucleotide similarity, at least 99.5% nucleotide similarity, at least 99.9% nucleotide similarity, or 10000 nucleotide similarity).

(78) TABLE-US-00008 TABLE 8 Representative Examples of Agmatinase by EMBL/GENBANK/DDBJ ID Number AAA24647.1 AAA69104.1 AAA83909.1 AAB98295.1 AAC75974.1 AAF40906.1 AAG58067.1 AAK40769.1 AAK42842.1 AAL21953.1 AAL51292.1 AAL52351.1 AAM30619.1 AAN44408.1 AAN67809.1 AAN81970.1 AAO70550.1 AAO90262.1 AAP12232.1 AAP18231.1 AAP29255.1 AAQ00893.1 AAQ58167.1 AAQ87313.1 AAQ87384.1 AAT34764.1 AAT57507.1 AAT63474.1 AAU15214.1 AAU45759.1 AAV44785.1 AAV46799.1 AAV78787.1 AAV93914.1 AAV95718.1 AAV95721.1 AAV97369.1 AAW87909.1 AAW90124.1 AAX66926.1 AAY90871.2 AAZ22133.1 AAZ24263.1 AAZ54096.1 AAZ58777.1 AAZ65050.1 AAZ69672.1 AAZ89678.1 ABA19753.1 ABA52140.1 ABA53789.1 ABA73200.1 ABA74317.1 ABA79070.1 ABA81349.1 ABB07975.1 ABB10482.1 ABB11261.1 ABB12898.1 ABB27188.1 ABB37375.1 ABB50840.1 ABB63153.1 ABB67558.1 ABC20115.1 ABC35193.1 ABC36254.1 ABC90370.1 ABC90927.1 ABC99063.1 ABD02701.1 ABD10232.1 ABD53194.1 ABD54884.1 ABD55257.1 ABD68100.1 ABD68348.1 ABE08773.1 ABE29546.1 ABE35008.1 ABE43955.1 ABE52778.1 ABE55558.1 ABE60184.1 ABE75327.1 ABF05058.1 ABF41119.1 ABF53436.1 ABF63671.1 ABF63672.1 ABF77979.1 ABF78588.1 ABG05817.1 ABG32253.1 ABG32638.1 ABG32639.1 ABG62558.1 ABG70915.1 ABG82607.1 ABG87141.1 ABI47535.1 ABI59683.1 AB160409.1 ABI67996.1 ABI89619.1 ABI90194.1 ABJ02365.1 ABK02249.1 ABK16419.1 ABK71425.1 ABK71845.1 ABK73126.1 ABK75016.1 ABK78003.1 ABK88044.1 ABL69259.1 ABL73176.1 ABL77915.1 ABL82726.1 ABL82883.1 ABL88073.1 ABL91072.1 ABL91501.1 ABL91972.1 ABL92277.1 ABM02702.1 ABM10897.1 ABM11012.1 ABM11652.1 ABM13827.1 ABM29529.1 ABM37937.1 ABM56124.1 ABM71179.1 ABM73084.1 ABM76715.1 ABM79681.1 ABM79929.1 ABM94848.1 ABM98665.2 ABN06601.1 ABN56898.1 ABN70230.1 ABN76664.1 ABN78567.1 ABN87189.1 ABN87725.1 ABN92921.1 ABN92962.1 ABO09043.1 ABO18500.1 ABO23488.1 ABO36121.1 ABO49035.1 ABO56523.1 ABO58149.1 ABP42590.1 ABP43281.1 ABP45780.1 ABP51374.1 ABP62005.1 ABP70012.1 ABP86798.1 ABP94673.1 ABP96350.1 ABQ18665.1 ABQ26532.1 ABQ31960.1 ABQ33008.1 ABQ60179.1 ABQ77546.1 ABQ79666.1 ABQ87081.1 ABQ91371.1 ABR78770.1 ABR81216.1 ABR81751.1 ABS77512.1 ABU75701.1 ABV07331.1 ABV15371.1 ABV21194.1 ABV43067.1 ABV51572.1 ABV67817.1 ABV93477.1 ABX09748.1 ABX69169.1 ABX74248.1 ABX81911.1 ABY21971.1 ACA38569.1 ACA76446.1 ACB04033.1 ACB18855.1 ACC84325.1 ACD06722.1 ACD24294.1 ACE90614.1 ACE91158.1 ACE94809.1 ACF30403.1 ACF61915.1 ACF67727.1 ACF91868.1 ACH50852.1 ACH63482.1 ACH77920.1 ACI10057.1 ACI11642.1 ACI18698.1 ACI35219.1 ACI78406.1 ACI78407.1 ACI78408.1 ACI78409.1 ACI78410.1 ACJ29740.1 ACK73593.1 ACM05018.1 ACM06157.1 ACN13261.1 ACN47228.1 ACO74132.1 ACP07733.1 ACP21851.1 ACP21922.1 ACP26294.1 ACP35785.1 ACP36686.1 ACR61943.1 ACR70510.1 ACS32933.1 ACS41939.1 ACS89313.1 ACU08695.1 ACV76715.1 ACV81107.1 ACY53515.1 ACY85795.1 ACY90127.1 ACZ87232.1 ADA75255.1 ADB74457.1 ADC90860.1 ADD45355.1 ADD58091.1 ADD78375.1 ADE04586.1 ADE37357.1 ADE39009.1 ADE40257.1 ADE40278.1 ADE67897.1 ADE69284.1 ADE72169.1 ADE85469.1 ADE86001.1 ADF37727.1 ADF39062.1 ADF41976.1 ADF63800.1 ADG91441.1

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(79) Representative examples of ornithine decarboxylase (EC 4.1.1.17) are given below in Table 9

and identified by their EMBL/GENBANK/DBJ ID numbers. Any of the bacteria given in Table 10 can be engineered with any version of the ornithine decarboxylase (EC 4.1.1.17) set forth in Table 3 and 9. For instance, the bacteria can be engineered with a version of the ornithine decarboxylase enzyme that has at least 50% nucleotide similarity with any of the versions of ornithine decarboxylase given in Table 9 (e.g., at least nucleotide 60% similarity, at least 70% nucleotide similarity, at least 80% nucleotide similarity, at least 90% nucleotide similarity, at least 91% nucleotide similarity, at least 92% nucleotide similarity, at least 93% nucleotide similarity, at least 94% nucleotide similarity, at least 95% nucleotide similarity, at least 96% nucleotide similarity, at least 97% nucleotide similarity, at least 98% nucleotide similarity, at least 99% nucleotide similarity, at least 99.5% nucleotide similarity, at least 99.9% nucleotide similarity, or 100% nucleotide similarity).

(80) TABLE-US-00009 TABLE 9 Representative Examples of Ornithine Decarboxylase by EMBL/GENBANK/DBJ ID Number AAA62785.1 AAN66489.2 ABB12186.1 ABE38094.1 AAA66174.1 AAN79250.1 ABB40870.1 ABE44334.1 AAA69133.1 AAN82001.1 ABB75259.1 ABE49183.1 AAC22248.1 AAO90264.2 ABC22493.1 ABE49327.1 AAC73787.1 AAS63636.1 ABC92384.1 ABE56353.1 AAC76002.2 AAV42846.1 ABD09230.1 ABE64149.1 AAK65018.1 AAW60223.1 ABD53718.1 ABE73821.1 AAK65019.2 AAY39277.1 ABD69850.1 ABF62084.1 AAL19645.1 AAY96264.1 ABD82224.1 ABF76724.1 AAL21989.1 AAZ17903.1 ABD90359.1 ABF79243.1 AAL54375.1 AAZ21026.1 ABD98955.1 ABG12272.1 AAL94697.1 AAZ45335.1 ABE05711.1 ABG19485.1 AAM32881.1 AAZ60071.1 ABE06188.1 ABG64092.1 AAM86897.1 AAZ71105.1 ABE08804.1 ABI25841.1 AAN53399.1 ABA04186.1 ABE29508.1 ABI41283.1 AAN57109.1 ABA78141.1 ABE36633.1 ABI41448.1 ABI59635.1 ABQ46942.1 ACF00856.1 ADG60285.1 ABI61420.1 ABQ77056.1 ACF62498.1 ADI02315.1 ABI64670.1 ABQ77881.1 ACF64187.1 ADM40347.1 ABI73241.1 ABQ94106.1 ACG58193.1 ADM42835.1 ABI88035.1 ABR63972.1 ACH49712.1 ADM42836.1 ABJ07102.1 ABR71097.1 ACH50260.1 ADM97227.1 ABJ08721.1 ABS65893.1 ACH63785.1 ADO10738.1 ABJ70427.1 ABS77454.2 ACI11630.1 ADO47055.1 ABK39746.1 ABV20152.1 ACI98761.1 ADO49370.1 ABK49901.1 ABV20217.1 ACI99115.1 ADO84265.1 ABK50089.1 ABV32717.1 ACJ00377.1 ADP99549.1 ABL98620.1 ABV39617.1 ACK48448.1 ADU45025.1 ABM01537.1 ABV43142.1 ACK49205.1 ADU70535.1 ABM03233.1 ABV93117.1 ACK81873.1 ADU91399.1 ABM17184.1 ABX51330.1 ACK83165.1 ADV11066.1 ABM33819.1 ABY29398.1 ACL55330.1 ADV56349.1 ABM37329.1 ABY96818.1 ACL93832.1 ADW75162.1 ABM42903.1 ACA15010.1 ACN14131.1 ADY12020.1 ABM94883.1 ACA32083.1 ACP07980.1 ADZ25852.1 ABN63325.1 ACA67112.1 ACP26670.1 ADZ25990.1 ABN63550.1 ACA74794.1 ACQ92481.1 AEA25536.1 ABO25258.1 ACA92668.1 ACR70548.1 AEA80789.1 ABO55514.1 ACB16558.1 ACR80274.1 AEB13218.1 ABP59886.1 ACB17513.1 ACS38696.1 AEE17405.1 ABP62042.1 ACB26370.1 ACS39829.1 AEF20700.1 ABP74273.1 ACB26621.1 ACS86807.1 AEF54819.1 ABP77387.1 ACB79112.1 ACT08032.1 AEG05601.1 ABP83569.1 ACB94577.1 ACT11922.1 AEG07021.1 ABQ13490.1 ACC75475.1 ACX89156.1 AEG09620.1 ABQ18377.1 ACD09708.1 ACY53777.1 AEG09845.1 ABQ31378.1 ACE89117.1 ACZ75768.1 AEG48476.1 ABQ33584.1 ACE92836.1 ADA66726.1 AEH50365.1 ABQ35247.1 ACE99495.1 ADE84258.1 AEH86445.1 AEI07722.1 AIJ07696.1 AKB72738.1 BAQ17945.1 AEJ18813.1 AIJ08459.1 AKB75272.1 BAU72598.1 AEK61463.1 AIQ92829.1 AKB78060.1 CAD78655.1 AEM41039.1 AIQ93098.1 AKB81641.1 CAD84820.1 AEP37160.1 AIR09930.1 AKB84150.1 CAG69512.1 AEV60919.1 AIT00368.1 AKF49823.1 CAG73878.1 AFJ59017.1 AIZ81713.1 AKH27073.1 CAH22470.1 AFL53867.1 AIZ84023.1 AKH69183.1 CAI97377.1 AFY17936.1 AJE57512.1 AKH99554.1 CAI98549.1 AFY20265.1 AJF55421.1 AKO77226.1 CAJ12264.1 AGA64911.1 AJF57769.1 AKO97889.1 CAK02040.1 AGG15787.1 AJG20615.1 ALD27842.1 CAK13937.1 AGG70048.1 AJI87080.1 ALV76207.1 CAL10580.1 AGG70049.1 AJK49051.1 ALY14469.1 CAL13549.1 AGH98733.1 AJQ96966.1 ALZ68204.1 CAL19626.1 AGL50812.1 AKA84438.1 ALZ69985.1 CAL78631.1 AGO56945.1 AKB12437.1 ANK35386.1

CAL79901.1 AGT07607.1 AKB14359.1 ANM83745.1 CAM00591.1 AHA66979.1 AKB18263.1
ANM85435.1 CAM03603.1 AHD10069.1 AKB21587.1 ANP16911.1 CAO97896.1 AHD10849.1
AKB25327.1 AOM43847.1 CAQ81820.1 AHE50504.1 AKB28151.1 AOT33749.1 CAQ89917.1
AHE58679.1 AKB32085.1 APF08471.1 CAQ90401.1 AHF70264.1 AKB37309.1 APF13004.1
CAR02067.1 AHF74955.1 AKB40448.1 BAA35349.1 CAR04482.1 AHH20127.1 AKB43558.1
BAE77028.1 CAR06878.1 AHJ62296.1 AKB47010.1 BAI71258.1 CAR09582.2 AHK70374.1
AKB51817.1 BAI72082.1 CAR14483.1 AHM04085.1 AKB55259.1 BAI76414.1 CAR16788.1
AHM04146.1 AKB56666.1 BAI96560.1 CAR19506.1 AHM77647.1 AKB61403.1 BAJ81775.1
CAR36594.1 AHM83238.1 AKB64701.1 BAL83346.1 CAR40820.1 AHY72323.1 AKB68292.1
BAN98715.1 CAR54968.1 CAU96547.1 CCD85313.1 CC007100.1 CDL50778.1 CAU99265.1
CCD87735.1 CCQ09200.1 CDL50819.1 CAX18544.1 CCD96638.1 CCQ74399.1 CDL86182.1
CAX22758.1 CCE00128.1 CCT59416.1 CDL90667.1 CAX24467.1 CCE04953.1 CCU62281.1
CDM25489.1 CAX61217.1 CCE05308.1 CCU62303.1 CDM39302.1 CAX67648.1 CCE97429.1
CCV05533.1 CDM59281.1 CAX67666.1 CCE98566.1 CCV16052.1 CDM60694.1 CAY75622.1
CCF20520.1 CCW14223.1 CDN08374.1 CBA19528.1 CCG07833.1 CCW18613.1 CDO59353.1
CBA28212.1 CCG39584.1 CDF57711.1 CDP53980.1 CBA29223.1 CCG86023.1 CDF82218.1
CDQ16217.1 CBA33757.1 CCG93567.1 CDF92784.1 CDR81373.1 CBG87473.1 CCI82390.1
CDF95827.1 CDR83974.1 CBG91695.1 CCJ43149.1 CDG34413.1 CDR89923.1 CBI76815.1
CCJ73160.1 CDG41242.1 CDS49197.1 CBI78256.1 CCJ73613.1 CDI09982.1 CDU34381.1
CBI78275.1 CCJ79388.1 CDK34386.1 CDU39548.1 CBL06633.1 CCJ82552.1 CDK72259.1
CDU41134.1 CBS85874.1 CCJ86161.1 CDK72653.1 CDX11380.1 CBS86483.1 CCJ87942.1
CDK75666.1 CDX28277.1 CBS89897.1 CCJ89861.1 CDK97758.1 CDX31971.1 CBW16782.1
CCJ91600.1 CDK98655.1 CDX32761.1 CBW19188.1 CCJ97363.1 CDL07277.1 CDX44074.1
CBY25820.1 CCJ99535.1 CDL10022.1 CDX54921.1 CBY25832.1 CCK01641.1 CDL20226.1
CDX58455.1 CBY28841.1 CCK03853.1 CDL23839.1 CDY74095.1 CCA93756.1 CCK06057.1
CDL25819.1 CEF30930.1 CCC57896.1 CCK08478.1 CDL25820.1 CEG53321.1 CCC96226.1
CCK22783.1 CDL27267.1 CEI15979.1 CCC98554.1 CCK45849.1 CDL27268.1 CEJ65390.1
CCD02602.1 CCK48255.1 CDL27269.1 CEK26252.1 CCD40282.1 CCM77346.1 CDL39170.1
CEK27948.1 CEL27520.1 CSC55030.1 CTS78438.1 EDQ31943.2 CEL88226.1 CSC90483.1
CTT62185.1 EDQ34501.1 CEP77894.1 CSC91956.1 CTT62210.1 EDR33413.1 CFQ53013.1
CSI90225.1 CTT77363.1 EDS92093.1 CFQ53087.1 CSI94697.1 CTT83136.1 EDU62245.1
CFQ53093.1 CSK70247.1 CTT84900.1 EDU64911.1 CFQ63135.1 CSL03024.1 CTT94762.1
EDU91579.1 CNJ42662.1 CSL07641.1 CTU09480.1 EDU92556.1 CNK09967.1 CSM63721.1
CTU43474.1 EDX33853.1 CPR14076.1 CSM98897.1 CTU46798.1 EDX36097.1 CPR26377.1
CSN54919.1 CTU67775.1 EDX47281.1 CPR26378.1 CSP82502.1 CTV17999.1 EDX48079.1
CQH28513.1 CSP97222.1 CTV22603.1 EDY22541.1 CQH50344.1 CSR81748.1 CTV49713.1
EDY26086.1 CRG52628.1 CSS46262.1 CTV56172.1 EDZ45245.1 CRH30711.1 CSS69716.1
CTV59366.1 EDZ48441.1 CRH35440.1 CST05596.1 CTV99708.1 EDZ60503.1 CRH39733.1
CTQ00658.1 CTW90830.1 EEB70922.1 CRL10245.1 CTQ07290.1 CTX56302.1 EEB72505.1
CRL11665.1 CTQ10899.1 CTZ36862.1 EEH90224.1 CRL87218.1 CTQ21239.1 CTZ54946.1
EEI47014.1 CRO16551.1 CTQ33367.1 CTZ68478.1 EEI67673.1 CRP80377.1 CTQ42919.1
CTZ90267.1 EEJ60488.1 CRY55295.1 CTQ43401.1 CUH38863.1 EEJ72107.1 CRY56151.1
CTQ48990.1 CUH97680.1 EEJ73250.1 CRY73398.1 CTQ51582.1 CUI46350.1 EEN80026.1
CRZ77491.1 CTQ64509.1 CUI65295.1 EE001571.1 CSA30444.1 CTQ70679.1 CUJ95194.1
EEO17726.1 CSB23271.1 CTQ77678.1 CUK08178.1 EEQ24327.1 CSB44423.1 CTR81841.1
EAA24580.1 EEV22915.1 CSB97832.1 CTR87483.1 EDL69175.1 EEW41165.1 CSC29054.1
CTS06036.1 EDM60163.1 EEX49266.1 CSC32042.1 CTS29676.1 EDN58782.1 EEY45602.1
EEY99726.1 EGK26884.1 E1Q71312.1 ENA38238.1 EFC91687.1 EGK34653.1 EIQ76994.1
ENA43866.1 EFE96168.1 EGK34656.1 EIQ78394.1 ENC90052.1 EFE96169.1 EGK40304.1
EIZ02840.1 END51232.1 EFE97489.1 EGK58241.1 EJI92011.1 END90160.1 EFG55431.1

EGK62838.1 EJL03608.1 ENG95671.1 EFM19932.1 EGL98455.1 EJJ08507.1 ENH01203.1 EFM60159.1 EGL98460.1 EJO49529.1 ENH07634.1 EFP71802.1 EGM68475.1 EK150690.1 EOD01056.1 EFQ23396.1 EGM68534.1 EKP99013.1 EPA96228.1 EFQ65259.1 EGO94943.1 EKP99014.1 EPC15663.1 EFR18105.1 EGP08625.1 EKQ02803.1 EPC38567.1 EFR18206.1 EGP13188.1 EKQ02804.1 EPP66737.1 EFW49207.1 EHC87901.1 EKQ09991.1 EPX75972.1 EFW49866.1 EHC94859.1 EKQ09992.1 EPX79893.1 EFW55944.1 EHC94874.1 EKQ11508.1 EPX85529.1 EFW57461.1 EHD22909.1 EKQ11509.1 EPX86372.1 EFW59125.1 EHE87725.1 EKQ18817.1 EQC66601.1 EFY04546.1 EHE90319.1 EKQ21433.1 EQC67119.1 EFZ35827.1 EHP88849.1 EKS52809.1 ERJ07090.1 EGC73182.1 EHP90237.1 ELS26426.1 ERJ12441.1 EGD26721.1 EHU02113.1 ELW31551.1 ERL16034.1 EGF12882.1 EHU20119.1 EMS72676.1 ERL56578.1 EGF12883.1 EHU26736.1 EMU59716.1 ESN64144.1 EGF13575.1 EHV55210.1 EMU68355.1 ESN64681.1 EGF13579.1 E1C02033.1 EMV17894.1 ESR23196.1 EGI92040.1 EIJ31771.1 EMW96757.1 ESR27061.1 EGI93253.1 EIK63890.1 EMX18650.1 ESS57441.1 EGI99407.1 EIK67948.1 EMX29506.1 ESU78166.1 EGJ01620.1 EIK70882.1 EMX36739.1 ETA69456.1 EGK15386.1 EIQ05587.1 EMX46897.1 ETS92400.1 EGK19303.1 EIQ18576.1 EMX83235.1 EUB20316.1 EGK26882.1 EIQ26095.1 ENA04683.1 EUB71374.1 EUB75197.1 KEJ45027.1 KGM62416.1 KMO67617.1 EUB84084.1 KEJ59156.1 KGM68326.1 KMO73993.1 EUB97023.1 KEJ73445.1 KGM71386.1 KMO74679.1 EUC12442.1 KEK27105.1 KGM77872.1 KMQ72910.1 EWC58917.1 KEK29881.1 KGM80580.1 KMY85028.1 EWC60810.1 KEL65139.1 KGM83330.1 KMZ12404.1 EWG73717.1 KEN53788.1 KGM84983.1 KND54388.1 EWG77348.1 KEN65519.1 KGM87536.1 KND59806.1 EYD71048.1 KEN98059.1 KGM88313.1 KNE76189.1 EYD71269.1 KEO08191.1 KHJ19976.1 KNF62494.1 EYD77132.1 KEO30074.1 KHN90703.1 KNF81913.1 EYD83397.1 KEQ53465.1 KHQ55244.1 KNF82765.1 EZJ36659.1 KER02947.1 KID27997.1 KNH02424.1 EZJ49393.1 KEY57491.1 KIH04277.1 KNH10085.1 EZJ71231.1 KEZ18534.1 KIQ45687.1 KNX39995.1 EZJ83500.1 KFC03280.1 KIQ47241.1 KNX42257.1 EZK19977.1 KFC05486.1 KIQ68453.1 KOA19269.1 EZP30980.1 KFC63691.1 KIQ68551.1 KON65543.1 EZP72267.1 KFC69417.1 KIR02360.1 KPA20904.1 GAO78552.1 KFC77644.1 KIR22223.1 KPA88263.1 GAP76203.1 KFC79295.1 KIS02883.1 KPO11674.1 KDA56789.1 KFC83942.1 KIT17643.1 KPO18606.1 KDE39162.1 KFC88765.1 KIV66814.1 KPO45157.1 KDM66230.1 KFC94984.1 KIV70876.1 KPO46605.1 KDM67094.1 KFD06768.1 KJL37417.1 KPP86129.1 KDM68717.1 KFD07842.1 KJL45196.1 KPQ30708.1 KDR44432.1 KFD17151.1 KJW30930.1 KPW02487.1 KDU29207.1 KFE39747.1 KJX87011.1 KPZ58476.1 KDW29935.1 KFE40588.1 KLU03251.1 KPZ73698.1 KDX21151.1 KFL92212.1 KLY03824.1 KQJ13961.1 KDX45737.1 KGA92942.1 KMK68540.1 KQJ37165.1 KEA62619.1 KGB23003.1 KMM41085.1 KRK54316.1 KEC54001.1 KGM59804.1 KMM41943.1 KRK64371.1 KSH09715.1 KYV67923.1 OCV53530.1 OJF06953.1 KST27486.1 KYV72148.1 ODH24445.1 SAY77725.1 KTZ13476.1 OAC43139.1 OEI63490.1 SEF59356.1 KXL65724.1 OAH24264.1 OEI65510.1 KXQ42753.1 OAJ89982.1 OFJ34340.1 KYR53760.1 OCS73051.1 OIR53362.1

(81) A variety of microbes (e.g., bacteria) can be engineered to produce GABA (e.g., by engineering one or more of the enzymes set forth in Table 2). For instance, any of the bacteria set forth in Table 10 can be engineered to produce GABA. In other words, a bacteria having a 16S rDNA nucleotide sequence that is at least 50% similar to that set forth in Table 10 below can be engineered (e.g., with one of the enzymes in Tables 3-9) to produce GABA. The bacteria may have a 16S rDNA sequence that is at least 60% similar, at least 70% similar, at least 80% similar, at least 90% similar, at least 91% similar, at least 92% similar, at least 93% similar, at least 94% similar, at least 95% similar, at least 96% similar, at least 97% similar, at least 98% similar, at least 99% similar, at least 99.5% similar, or 100% similar to the 16S rDNA nucleotide sequence given in Table 10.

(82) As set forth in Example 8, *E. coli* was engineered to overexpress glutamate decarboxylase, which without wishing to be bound by theory, led to the expression of GABA by the engineered *E.*

coli. As shown in FIGS. 6A-E, the engineered *E. coli* was able to induce the growth of *E. gabavorous* KLE1738. GABA can be produced by intestinal epithelial cells and by some bacteria, such as *Escherichia coli* and *Listeria monocytogenes*, by decarboxylation of glutamate. In *E. coli*, the decarboxylation of glutamate can serve as a mechanism to decrease intracellular pH, and therefore in some embodiments GABA production generally occurs at a low pH.

(83) As set forth in Example 8, Expression, e.g., overexpression of glutamate decarboxylase in *E. coli* (e.g., *gadA* or *gadB*), resulted in induction of *E. gabavorous* KLE1738 growth to levels seen with *B. fragilis* KLE 1758. For instance, FIG. 6A shows induction of *E. gabavorous* KLE1738 in the presence of *E. coli* engineered to express glutamate decarboxylase *gadA*. Similarly, FIG. 6B shows induction of *E. gabavorous* KLE1738 in the presence of *E. coli* engineered to express glutamate decarboxylase *gadB*. As shown in FIG. 6C, the growth of *E. gabavorous* KLE1738 was qualitatively similar to that seen in the presence of *B. fragilis* KLE 1758. In contrast, as shown in FIG. 6D, when *E. coli* was engineered to express *gadC*, a GABA antiporter, no growth of *E. gabavorous* KLE1738 was observed. Similarly, no growth of *E. gabavorous* KLE1738 was observed in the presence of empty vehicle (FIG. 6E). Without wishing to be bound by theory the results of Example 8 demonstrate that bacteria can be engineered to produce GABA (e.g., via the expression or overexpression of glutamate decarboxylase). In some embodiments, the bacteria (e.g., *E. coli*) can be engineered to produce GABA inside the human gut.

(84) In some embodiments, the present disclosure also provides compromising one or more repressors of GABA production, (e.g., *gadX* or *gadW*). In some embodiments, these repressors can regulate the pH restrictions of GABA production in *E. coli*, is a way to increase native GABA production. This can be achieved, for instance, via gene deletions, insertions, or substitutions, as known by those skilled in the art of molecular biology.

(85) Altering the pH of growth media for KLE1738 did not change the GABA-dependency phenotype. Without wishing to be bound by theory, this suggests that engineering bacteria to overexpress glutamate decarboxylase is an effective way to produce GABA, as well as induce the growth of *E. gabavorous*.

(86) In addition to *E. coli*, other bacteria can be engineered to produce GABA (e.g., at a physiologically relevant pH, such as between 4.5 and 7.5). For instance, any of the bacteria set forth in Table 10 can be engineered to produce GABA (e.g., at a physiologically relevant pH, such as between 4.5 and 7.5). For example, the bacteria can be engineered to contain DNA that codes for one or more enzymes set forth in Table 3-9. Set forth in Table 10 is also a sequence ID number for the 16S nucleotide sequence of the listed bacteria. In some embodiments, the bacteria that are engineered to produce GABA can have at least 90% 16S sequence similarity to the 16S sequences given in Table 10 (e.g., at least 91% similarity, at least 92% similarity, at least 93% similarity, at least 94% similarity, at least 95% similarity, at least 96% similarity, at least 97% similarity, at least 98% similarity, at least 99% similarity, at least 99.5% similarity, at least 99.9% similarity, or 100% similarity).

(87) TABLE-US-00010 TABLE 10 Bacteria Capable of Being Engineered to Produce GABA

Species	Seq. ID No.
<i>Abiotrophia defectiva</i>	305
<i>Abiotrophia para-adiacens</i>	306
<i>Abiotrophia</i> sp.	307
<i>Acetanaerobacterium elongatum</i>	308
<i>Acetivibrio cellulolyticus</i>	309
<i>Acetivibrio ethanolgignens</i>	310
<i>Acetobacter aceti</i>	311
<i>Acetobacter fabarum</i>	312
<i>Acetobacter lovaniensis</i>	313
<i>Acetobacter malorum</i>	314
<i>Acetobacter orientalis</i>	315
<i>Acetobacter pasteurianus</i>	316
<i>Acetobacter pomorum</i>	317
<i>Acetobacter syzygii</i>	318
<i>Acetobacter tropicalis</i>	319
<i>Acetobacteraceae bacterium</i>	320
<i>Acholeplasma laidlawii</i>	321
<i>Achromobacter denitrificans</i>	322
<i>Achromobacter piechaudii</i>	323
<i>Achromobacter xylosoxidans</i>	324
<i>Acidaminococcus fermentans</i>	325
<i>Acidaminococcus intestini</i>	326
<i>Acidaminococcus</i> sp.	327
<i>Acidilobus saccharovorans</i>	328
<i>Acidithiobacillus ferrivorans</i>	329
<i>Acidovorax</i> sp.	330
<i>Acidovorax</i> sp.	331
<i>Acinetobacter baumannii</i>	332
<i>Acinetobacter calcoaceticus</i>	333
<i>Acinetobacter genomos</i> sp.	334
<i>Acinetobacter haemolyticus</i>	335
<i>Acinetobacter johnsonii</i>	336
<i>Acinetobacter junii</i>	337
<i>Acinetobacter lwoffii</i>	338
<i>Acinetobacter parvus</i>	339
<i>Acinetobacter</i>	

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Gluconacetobacter azotocaptans 1175 *Gluconacetobactercdiazotrophicus* 1176 *Gluconacetobacter*
entanii 1177 *Gluconacetobacter europaeus* 1178 *Gluconacetobacter hansenii* 1179
Gluconacetobacter johannae 1180 *Gluconacetobacter oboediens* 1181 *Gluconacetobacter xylinus* 1182
Gordonia bronchialis 1183 *Gordonia polyisoprenivorans* 1184 *Gordonia* sp. 1185 *Gordonia sputi* 1186
Gordonia terrae 1187 *Gordonibacter pamelaee* 1188 *Gordonibacter pamelaee* 1189
Gracilibacter thermotolerans 1190 *Gramella forsetii* 1191 *Granulicatella adiacens* 1192
Granulicatella elegans 1193 *Granulicatella paradiacens* 1194 *Granulicatella* sp. 1195
Granulicatella sp. 1196 *Granulicatella* sp. 1197 *Granulicatella* sp. 1198 *Granulicatella* sp. 1199
Grimontia hollisae 1200 *Haematobacter* sp. 1201 *Haemophilus aegyptius* 1202 *Haemophilus*

genomo sp. 1203 *Haemophilus* genomo sp. 1204 *Haemophilus* haemolyticus 1205 *Haemophilus* parahaemolyticus 1206 *Haemophilus* parainfluenzae 1207 *Haemophilus* paraphrophaemolyticus 1208 *Haemophilus* parasuis 1209 *Haemophilus* somnus 1210 *Haemophilus* sp. 1211 *Haemophilus* sp. 1212 *Haemophilus* sp. 1213 *Haemophilus* sp. 1214 *Haemophilus* sp. 1215 *Haemophilus* sp. 1216 *Haemophilus* sp. 1217 *Haemophilus* sp. 1218 *Haemophilus* sputorum 1219 *Hafnia* alvei 1220 *Halomonas* elongata 1221 *Halomonas* johnsoniae 1222 *Halorubrum* lipolyticum 1223 *Helicobacter* bilis 1224 *Helicobacter* canadensis 1225 *Helicobacter* cinaedi 1226 *Helicobacter* pullorum 1227 *Helicobacter* sp. 1228 *Helicobacter* winthamensis 1229 *Heliobacterium* modesticaldum 1230 *Herbaspirillum* seropedicae 1231 *Herbaspirillum* sp. 1232 *Histophilus* somni 1233 *Holdemania* filiformis 1234 *Hydrogenoanaerobacterium* saccharovorans 1235 *Hyperthermus* butylicus 1236 *Hyphomicrobium* sulfonivorans 1237 *Hyphomonas* neptunium 1238 *Ignatzschineria* indica 1239 *Ignatzschineria* sp. 1240 *Ignicoccus* islandicus 1241 *Inquilinus* limosus 1242 *Janibacter* limosus 1243 *Janibacter* melonis 1244 *Janthinobacterium* sp. 1245 *Johnsonella* ignava 1246 *Jonquetella* anthropi 1247 *Kerstesia* gyiorum 1248 *Kingella* denitrificans 1249 *Kingella* genomo sp. 1250 *Kingella* kingae 1251 *Kingella* oralis 1252 *Kingella* sp. 1253 *Klebsiella* sp. 1254 *Klebsiella* sp. 1255 *Klebsiella* sp. 1256 *Klebsiella* sp. 1257 *Klebsiella* sp. 1258 *Klebsiella* sp. 1259 *Klebsiella* sp. 1260 *Klebsiella* sp. 1261 *Klebsiella* sp. 1262 *Klebsiella* sp. 1263 *Klebsiella* sp. 1264 *Klebsiella* variicola 1265 *Kluyvera* ascorbata 1266 *Kluyvera* cryocrescens 1267 *Kocuria* marina 1268 *Kocuria* palustris 1269 *Kocuria* rhizophila 1270 *Kocuria* rosea 1271 *Kocuria* varians 1272 *Lachnobacterium* bovis 1273 *Lachnospira* multipara 1274 *Lachnospira* pectinoschiza 1275 *Lachnospiraceae* bacterium 1276 *Lachnospiraceae* bacterium 1277 *Lachnospiraceae* bacterium 1278 *Lachnospiraceae* bacterium 1279 *Lachnospiraceae* bacterium 1280 *Lachnospiraceae* bacterium 1281 *Lachnospiraceae* bacterium 1282 *Lachnospiraceae* bacterium 1283 *Lachnospiraceae* bacterium 1284 *Lachnospiraceae* bacterium 1285 *Lachnospiraceae* bacterium 1286 *Lachnospiraceae* bacterium 1287 *Lachnospiraceae* bacterium 1288 *Lachnospiraceae* bacterium 1289 *Lachnospiraceae* bacterium 1290 *Lachnospiraceae* bacterium 1291 *Lachnospiraceae* bacterium 1292 *Lachnospiraceae* genomo sp. 1293 *Lactobacillus* acidipiscis 1294 *Lactobacillus* acidophilus 1295 *Lactobacillus* alimentarius 1296 *Lactobacillus* amylolyticus 1297 *Lactobacillus* amylovorus 1298 *Lactobacillus* antri 1299 *Lactobacillus* brevis 1300 *Lactobacillus* buchneri 1301 *Lactobacillus* casei 1302 *Lactobacillus* catenaformis 1303 *Lactobacillus* coleohominis 1304 *Lactobacillus* coryniformis 1305 *Lactobacillus* crispatus 1306 *Lactobacillus* curvatus 1307 *Lactobacillus* delbrueckii 1308 *Lactobacillus* dextrinicus 1309 *Lactobacillus* farciminis 1310 *Lactobacillus* fermentum 1311 *Lactobacillus* gasseri 1312 *Lactobacillus* gastricus 1313 *Lactobacillus* genomo sp. 1314 *Lactobacillus* genomo sp. 1315 *Lactobacillus* helveticus 1316 *Lactobacillus* hilgardii 1317 *Lactobacillus* hominis 1318 *Lactobacillus* iners 1319 *Lactobacillus* jensenii 1320 *Lactobacillus* johnsonii 1321 *Lactobacillus* kalixensis 1322 *Lactobacillus* kefiranoferiens 1323 *Lactobacillus* kefir 1324 *Lactobacillus* kimchii 1325 *Lactobacillus* leichmannii 1326 *Lactobacillus* mucosae 1327 *Lactobacillus* murinus 1328 *Lactobacillus* nodensis 1329 *Lactobacillus* oeni 1330 *Lactobacillus* oris 1331 *Lactobacillus* parabrevis 1332 *Lactobacillus* parabuchneri 1333 *Lactobacillus* paracasei 1334 *Lactobacillus* parakefir 1335 *Lactobacillus* pentosus 1336 *Lactobacillus* perolens 1337 *Lactobacillus* plantarum 1338 *Lactobacillus* pontis 1339 *Lactobacillus* reuteri 1340 *Lactobacillus* rhamnosus 1341 *Lactobacillus* rogosae 1342 *Lactobacillus* ruminis 1343 *Lactobacillus* sakei 1344 *Lactobacillus* salivarius 1345 *Lactobacillus* saniviri 1346 *Lactobacillus* senioris 1347 *Lactobacillus* sp. 1348 *Lactobacillus* sp. 1349 *Lactobacillus* sp. 1350 *Lactobacillus* sp. 1351 *Lactobacillus* sp. 1352 *Lactobacillus* sp. 1353 *Lactobacillus* sp. 1354 *Lactobacillus* sp. 1355 *Lactobacillus* sp. 1356 *Lactobacillus* sp. 1357 *Lactobacillus* sp. 1358 *Lactobacillus* sp. 1359 *Lactobacillus* sp. 1360 *Lactobacillus* sp. 1361 *Lactobacillus* sp. 1362 *Lactobacillus* sp. 1363 *Lactobacillus* sp. 1364 *Lactobacillus* sp. 1365 *Lactobacillus* tucseti 1366 *Lactobacillus* ultunensis 1367 *Lactobacillus* vaginalis 1368 *Lactobacillus* vini 1369 *Lactobacillus* vitulinus 1370 *Lactobacillus* zeae 1371

Lactococcus garvieae 1372 *Lactococcus lactis* 1373 *Lactococcus raffinolactis* 1374 *Lactonifactor longoviformis* 1375 *Laribacter hongkongensis* 1376 *Lautropia mirabilis* 1377 *Lautropia* sp. 1378 *Leminorella grimontii* 1379 *Leminorella richardii* 1380 *Leptotrichia buccalis* 1381 *Leptotrichia genomo* sp. 1382 *Leptotrichia goodfellowii* 1383 *Leptotrichia hofstadii* 1384 *Leptotrichia shahii* 1385 *Leptotrichia* sp. 1386 *Leptotrichia* sp. 1387 *Leptotrichia* sp. 1388 *Leptotrichia* sp. 1389 *Leptotrichia* sp. 1390 *Leptotrichia* sp. 1391 *Leptotrichia* sp. 1392 *Leuconostoc carnosum* 1393 *Leuconostoc citreum* 1394 *Leuconostoc gasicomitatum* 1395 *Leuconostoc inhae* 1396 *Leuconostoc kimchii* 1397 *Leuconostoc lactis* 1398 *Leuconostoc mesenteroides* 1399 *Leuconostoc pseudomesenteroides* 1400 *Listeria innocua* 1401 *Listeria ivanovii* 1402 *Luteococcus sanguinis* 1403 *Lutispora thermophila* 1404 *Lysinibacillus fusiformis* 1405 *Lysinibacillus sphaericus* 1406 *Macrococcus caseolyticus* 1407 *Mannheimia haemolytica* 1408 *Marvinbryantia formatexigens* 1409 *Massilia* sp. 1410 *Megamonas funiformis* 1411 *Megamonas hypermegale* 1412 *Megasphaera genomo* sp. 1413 *Megasphaera genomo* sp. 1414 *Megasphaera micronuciformis* 1415 *Megasphaera* sp. 1416 *Megasphaera* sp. 1417 *Metallosphaera sedula* 1418 *Methanobacterium formicicum* 1419 *Methanobrevibacter acididurans* 1420 *Methanobrevibacter arboriphilus* 1421 *Methanobrevibacter curvatus* 1422 *Methanobrevibacter cuticularis* 1423 *Methanobrevibacter filiformis* 1424 *Methanobrevibacter gottschalkii* 1425 *Methanobrevibacter millerae* 1426 *Methanobrevibacter olleyae* 1427 *Methanobrevibacter orails* 1428 *Methanobrevibacter ruminantium* 1429 *Methanobrevibacter smithii* 1430 *Methanobrevibacter thaueri* 1431 *Methanobrevibacter woesei* 1432 *Methanobrevibacter wolinii* 1433 *Methanosphaera stadtmanae* 1434 *Methylobacterium extorquens* 1435 *Methylobacterium podarium* 1436 *Methylobacterium radiotolerans* 1437 *Methylobacterium* sp. 1438 *Methylobacterium* sp. 1439 *Methylocella silvestris* 1440 *Methylophilus* sp. 1441 *Microbacterium chocolatum* 1442 *Microbacterium flavescens* 1443 *Microbacterium gubbeenense* 1444 *Microbacterium lacticum* 1445 *Microbacterium oleivorans* 1446 *Microbacterium oxydans* 1447 *Microbacterium paraoxydans* 1448 *Microbacterium phyllosphaerae* 1449 *Microbacterium schleiferi* 1450 *Microbacterium* sp. 1451 *Microbacterium* sp. 1452 *Microbacterium testaceum* 1453 *Micrococcus antarcticus* 1454 *Micrococcus luteus* 1455 *Micrococcus lylae* 1456 *Micrococcus* sp. 1457 *Microcystis aeruginosa* 1458 *Mitsuokella jalaludinii* 1459 *Mitsuokella multacida* 1460 *Mitsuokella* sp. 1461 *Mitsuokella* sp. 1462 *Mobiluncus curtisii* 1463 *Mobiluncus mulieris* 1464 *Moellerella wisconsensis* 1465 *Mogibacterium diversum* 1466 *Mogibacterium neglectum* 1467 *Mogibacterium pumilum* 1468 *Mogibacterium timidum* 1469 *Mollicutes bacterium* 1470 *Moorella thermoacetica* 1471 *Moraxella catarrhalis* 1472 *Moraxella lincolnii* 1473 *Moraxella osloensis* 1474 *Moraxella* sp. 1475 *Moraxella* sp. 1476 *Morganella morganii* 1477 *Morganella* sp. 1478 *Morococcus cerebrosus* 1479 *Moryella indoligenes* 1480 *Mycobacterium* sp. 1481 *Mycobacterium* sp. 1482 *Mycobacterium* sp. 1483 *Mycobacterium* sp. 1484 *Mycobacterium* sp. 1485 *Mycobacterium* sp. 1486 *Mycobacterium* sp. 1487 *Mycobacterium* sp. 1488 *Mycobacterium* sp. 1489 *Mycobacterium* sp. 1490 *Mycobacterium* sp. 1491 *Mycobacterium* sp. 1492 *Mycobacterium* sp. 1493 *Mycobacterium* sp. 1494 *Mycobacterium* sp. 1495 *Mycoplasma agalactiae* 1496 *Mycoplasma amphoriforme* 1497 *Mycoplasma arthritidis* 1498 *Mycoplasma bovoculi* 1499 *Mycoplasma faucium* 1500 *Mycoplasma fermentans* 1501 *Mycoplasma flocculare* 1502 *Mycoplasma genitalium* 1503 *Mycoplasma hominis* 1504 *Mycoplasma orale* 1505 *Mycoplasma ovipneumoniae* 1506 *Mycoplasma penetrans* 1507 *Mycoplasma pneumoniae* 1508 *Mycoplasma putrefaciens* 1509 *Mycoplasma salivarium* 1510 *Mycoplasmataceae genomo* sp. 1511 *Myroides odoratimimus* 1512 *Myroides* sp. 1513 *Neisseria bacilliformis* 1514 *Neisseria cinerea* 1515 *Neisseria elongata* 1516 *Neisseria flavescens* 1517 *Neisseria genomo* sp. 1518 *Neisseria lactamica* 1519 *Neisseria macacae* 1520 *Neisseria mucosa* 1521 *Neisseria pharyngis* 1522 *Neisseria polysaccharea* 1523 *Neisseria sicca* 1524 *Neisseria* sp. 1525 *Neisseria* sp. 1526 *Neisseria* sp. 1527 *Neisseria* sp. 1528 *Neisseria* sp. 1529 *Neisseria* sp. 1530 *Neisseria* sp. 1531 *Neisseria subflava* 1532 *Neorickettsia risticii* 1533 *Neorickettsia sennetsu* 1534 *Nocardia brasiliensis* 1535 *Nocardia cyriacigeorgica* 1536 *Nocardia farcinica* 1537 *Nocardia purlis* 1538

Nocardia sp. 1539 *Nocardiopsis dassonvillei* 1540 *Novosphingobium aromaticivorans* 1541
Oceanobacillus caeni 1542 *Oceanobacillus* sp. 1543 *Ochrobactrum anthropi* 1544 *Ochrobactrum*
intermedium 1545 *Ochrobactrum pseudintermedium* 1546 *Odoribacter laneus* 1547 *Odoribacter*
splanchnicus 1548 *Okadaella gastrococcus* 1549 *Oligella ureolytica* 1550 *Oligella urethralis* 1551
Olsenella genomus sp. 1552 *Olsenella profusa* 1553 *Olsenella* sp. 1554 *Olsenella* sp. 1555
Olsenella uli 1556 *Opitutus terrae* 1557 *Oribacterium sinus* 1558 *Oribacterium* sp. 1559
Oribacterium sp. 1560 *Oribacterium* sp. 1561 *Oribacterium* sp. 1562 *Oribacterium* sp. 1563
Oribacterium sp. 1564 *Oribacterium* sp. 1565 *Oribacterium* sp. 1566 *Ornithinibacillus bavariensis*
1567 *Ornithinibacillus* sp. 1568 *Oscillibacter* sp. 1569 *Oscillibacter valericigenes* 1570
Oscillospira guilliermondii 1571 *Oxalobacter formigenes* 1572 *Paenibacillus barcinonensis* 1573
Paenibacillus barengoltzii 1574 *Paenibacillus chibensis* 1575 *Paenibacillus cookii* 1576
Paenibacillus durus 1577 *Paenibacillus glucanolyticus* 1578 *Paenibacillus lactis* 1579
Paenibacillus lautus 1580 *Paenibacillus pabuli* 1581 *Paenibacillus polymyxa* 1582 *Paenibacillus*
popilliae 1583 *Paenibacillus* sp. 1584 *Paenibacillus* sp. 1585 *Paenibacillus* sp. 1586 *Paenibacillus*
sp. 1587 *Paenibacillus* sp. 1588 *Paenibacillus* sp. 1589 *Paenibacillus* sp. 1590 *Paenibacillus*
timonensis 1591 *Pantoea agglomerans* 1592 *Pantoea ananatis* 1593 *Pantoea brenneri* 1594
Pantoea citrea 1595 *Pantoea conspicua* 1596 *Pantoea septica* 1597 *Papillibacter cinnamivorans*
1598 *Parabacteroides distasonis* 1599 *Parabacteroides goldsteinii* 1600 *Parabacteroides gordonii*
1601 *Parabacteroides johnsonii* 1602 *Parabacteroides merdae* 1603 *Parabacteroides* sp. 1604
Parabacteroides sp. 1605 *Parachlamydia* sp. 1606 *Paracoccus denitrificans* 1607 *Paracoccus*
marcusii 1608 *Paraprevotella clara* 1609 *Paraprevotella xylaniphila* 1610 *Parascardovia*
denticolens 1611 *Parasutterella excrementihominis* 1612 *Parasutterella secunda* 1613 *Parvimonas*
micra 1614 *Parvimonas* sp. 1615 *Pasteurella bettyae* 1616 *Pasteurella dagmatis* 1617 *Pasteurella*
multocida 1618 *Pediococcus acidilactici* 1619 *Pediococcus pentosaceus* 1620 *Peptococcus niger*
1621 *Peptococcus* sp. 1622 *Peptococcus* sp. 1623 *Peptoniphilus asaccharolyticus* 1624
Peptoniphilus duerdenii 1625 *Peptoniphilus harei* 1626 *Peptoniphilus indolicus* 1627
Peptoniphilus ivorii 1628 *Peptoniphilus lacrimalis* 1629 *Peptoniphilus* sp. 1630 *Peptoniphilus* sp.
1631 *Peptoniphilus* sp. 1632 *Peptoniphilus* sp. 1633 *Peptoniphilus* sp. 1634 *Peptoniphilus* sp. 1635
Peptoniphilus sp. 1636 *Peptostreptococcaceae bacterium* 1637 *Peptostreptococcus anaerobius*
1638 *Peptostreptococcus micros* 1639 *Peptostreptococcus* sp. 1640 *Peptostreptococcus* sp. 1641
Peptostreptococcus sp. 1642 *Peptostreptococcus* sp. 1643 *Peptostreptococcus stomatis* 1644
Phascolarctobacterium faecium 1645 *Phascolarctobacterium* sp. 1646 *Phascolarctobacterium*
succinatutens 1647 *Phenylobacterium zucineum* 1648 *Photorhabdus asymbiotica* 1649
Pigmentiphaga daeguensis 1650 *Planomicrobium koreense* 1651 *Plesiomonas shigelloides* 1652
Porphyromonadaceae bacterium 1653 *Porphyromonas asaccharolytica* 1654 *Porphyromonas*
endodontalis 1655 *Porphyromonas gingivalis* 1656 *Porphyromonas levii* 1657 *Porphyromonas*
macacae 1658 *Porphyromonas somerae* 1659 *Porphyromonas* sp. 1660 *Porphyromonas* sp. 1661
Porphyromonas sp. 1662 *Porphyromonas* sp. 1663 *Porphyromonas* sp. 1664 *Porphyromonas*
uenonis 1665 *Prevotella albensis* 1666 *Prevotella amnii* 1667 *Prevotella bergensis* 1668 *Prevotella*
bivia 1669 *Prevotella brevis* 1670 *Prevotella buccae* 1671 *Prevotella buccalis* 1672 *Prevotella*
copri 1673 *Prevotella corporis* 1674 *Prevotella dentalis* 1675 *Prevotella denticola* 1676 *Prevotella*
disiens 1677 *Prevotella genomus* sp. 1678 *Prevotella genomus* sp. 1679 *Prevotella genomus* sp. 1680
Prevotella genomus sp. 1681 *Prevotella genomus* sp. 1682 *Prevotella heparinolytica* 1683 *Prevotella*
histicola 1684 *Prevotella intermedia* 1685 *Prevotella loescheii* 1686 *Prevotella maculosa* 1687
Prevotella marshii 1688 *Prevotella melaninogenica* 1689 *Prevotella micans* 1690 *Prevotella*
multiformis 1691 *Prevotella multisaccharivora* 1692 *Prevotella nanceiensis* 1693 *Prevotella*
nigrescens 1694 *Prevotella oralis* 1695 *Prevotella oris* 1696 *Prevotella oulorum* 1697 *Prevotella*
pallens 1698 *Prevotella ruminicola* 1699 *Prevotella salivae* 1700 *Prevotella* sp. 1701 *Prevotella* sp.
1702 *Prevotella* sp. 1703 *Prevotella* sp. 1704 *Prevotella* sp. 1705 *Prevotella* sp. 1706 *Prevotella* sp.
1707 *Prevotella* sp. 1708 *Prevotella* sp. 1709 *Prevotella* sp. 1710 *Prevotella* sp. 1711 *Prevotella* sp.

1712 *Prevotella* sp. 1713 *Prevotella* sp. 1714 *Prevotella* sp. 1715 *Prevotella* sp. 1716 *Prevotella* sp.
1717 *Prevotella* sp. 1718 *Prevotella* sp. 1719 *Prevotella* sp. 1720 *Prevotella* sp. 1721 *Prevotella* sp.
1722 *Prevotella* sp. 1723 *Prevotella* sp. 1724 *Prevotella* sp. 1725 *Prevotella* sp. 1726 *Prevotella* sp.
1727 *Prevotella* sp. 1728 *Prevotella* sp. 1729 *Prevotella* sp. 1730 *Prevotella* sp. 1731 *Prevotella* sp.
1732 *Prevotella* sp. 1733 *Prevotella* sp. 1734 *Prevotella* sp. 1735 *Prevotella* sp. 1736 *Prevotella* sp.
1737 *Prevotella* sp. 1738 *Prevotella* sp. 1739 *Prevotella* sp. 1740 *Prevotella* sp. 1741 *Prevotella* sp.
1742 *Prevotella* sp. 1743 *Prevotella* sp. 1744 *Prevotella* sp. 1745 *Prevotella* sp. 1746 *Prevotella*
stercorea 1747 *Prevotella tanneriae* 1748 *Prevotella timonensis* 1749 *Prevotella veroralis* 1750
Prevotellaceae bacterium 1751 *Prochlorococcus marinus* 1752 *Propionibacteriaceae* bacterium
1753 *Propionibacterium acidipropionici* 1754 *Propionibacterium acnes* 1755 *Propionibacterium*
avidum 1756 *Propionibacterium freudenreichii* 1757 *Propionibacterium granulosum* 1758
Propionibacterium jensenii 1759 *Propionibacterium propionicum* 1760 *Propionibacterium* sp.
1761 *Propionibacterium* sp. 1762 *Propionibacterium* sp. 1763 *Propionibacterium* sp. 1764
Propionibacterium sp. 1765 *Propionibacterium thoenii* 1766 *Proteus mirabilis* 1767 *Proteus*
pennneri 1768 *Proteus* sp. 1769 *Proteus vulgaris* 1770 *Providencia alcalifaciens* 1771 *Providencia*
rettgeri 1772 *Providencia rustigianii* 1773 *Providencia stuartii* 1774 *Pseudoclavibacter* sp. 1775
Pseudoflavonifractor capillosus 1776 *Pseudomonas aeruginosa* 1777 *Pseudomonas fluorescens*
1778 *Pseudomonas gessardii* 1779 *Pseudomonas mendocina* 1780 *Pseudomonas monteilii* 1781
Pseudomonas poae 1782 *Pseudomonas pseudoalcaligenes* 1783 *Pseudomonas putida* 1784
Pseudomonas sp. 1785 *Pseudomonas* sp. 1786 *Pseudomonas* sp. 1787 *Pseudomonas stutzeri* 1788
Pseudomonas tolaasii 1789 *Pseudomonas viridiflava* 1790 *Pseudoramibacter alactolyticus* 1791
Psychrobacter arcticus 1792 *Psychrobacter cibarius* 1793 *Psychrobacter cryohalolentis* 1794
Psychrobacter faecalis 1795 *Psychrobacter nivimaris* 1796 *Psychrobacter pulmonis* 1797
Psychrobacter sp. 1798 *Pyramidobacter piscolens* 1799 *Ralstonia pickettii* 1800 *Ralstonia* sp.
1801 *Raoultella ornithinolytica* 1802 *Raoultella planticola* 1803 *Raoultella terrigena* 1804
Rhodobacter sp. 1805 *Rhodobacter sphaeroides* 1806 *Rhodococcus corynebacterioides* 1807
Rhodococcus equi 1808 *Rhodococcus erythropolis* 1809 *Rhodococcus fascians* 1810
Rhodopseudomonas palustris 1811 *Robinsoniella peoriensis* 1812 *Roseburia cecicola* 1813
Roseburia faecalis 1814 *Roseburia faecis* 1815 *Roseburia hominis* 1816 *Roseburia intestinalis*
1817 *Roseburia inulinivorans* 1818 *Roseburia* sp. 1819 *Roseburia* sp. 1820 *Roseiflexus*
castenholzii 1821 *Roseomonas cervicalis* 1822 *Roseomonas mucosa* 1823 *Roseomonas* sp. 1824
Roseomonas sp. 1825 *Roseomonas* sp. 1826 *Roseomonas* sp. 1827 *Rothia aerea* 1828 *Rothia*
dentocariosa 1829 *Rothia mucilaginoso* 1830 *Rothia nasimurium* 1831 *Rothia* sp. 1832
Ruminobacter amylophilus 1833 *Ruminococcaceae* bacterium 1834 *Ruminococcus albus* 1835
Ruminococcus bromii 1836 *Ruminococcus callidus* 1837 *Ruminococcus champanellensis* 1838
Ruminococcus flavefaciens 1839 *Ruminococcus gnavus* 1840 *Ruminococcus hansenii* 1841
Ruminococcus lactaris 1842 *Ruminococcus obeum* 1843 *Ruminococcus* sp. 1844 *Ruminococcus* sp.
1845 *Ruminococcus* sp. 1846 *Ruminococcus* sp. 1847 *Ruminococcus* sp. 1848 *Ruminococcus*
torques 1849 *Saccharomonospora viridis* 1850 *Sarcina ventriculi* 1851 *Scardovia inopinata* 1852
Scardovia wiggisiae 1853 *Segniliparus rotundus* 1854 *Segniliparus rugosus* 1855 *Selenomonas*
artemidis 1856 *Selenomonas diana* 1857 *Selenomonas flueggei* 1858 *Selenomonas genom* sp.
1859 *Selenomonas genom* sp. 1860 *Selenomonas genom* sp. 1861 *Selenomonas genom* sp. 1862
Selenomonas genom sp. 1863 *Selenomonas genom* sp. 1864 *Selenomonas infeli* 1865
Selenomonas noxia 1866 *Selenomonas ruminantium* 1867 *Selenomonas* sp. 1868 *Selenomonas* sp.
1869 *Selenomonas* sp. 1870 *Selenomonas* sp. 1871 *Selenomonas* sp. 1872 *Selenomonas* sp. 1873
Selenomonas sp. 1874 *Selenomonas* sp. 1875 *Selenomonas* sp. 1876 *Selenomonas* sp. 1877
Selenomonas sp. 1878 *Selenomonas* sp. 1879 *Selenomonas* sp. 1880 *Selenomonas sputigena* 1881
Serratia fonticola 1882 *Serratia liquefaciens* 1883 *Serratia marcescens* 1884 *Serratia odorifera*
1885 *Serratia proteamaculans* 1886 *Shewanella putrefaciens* 1887 *Shuttleworthia satellites* 1888
Shuttleworthia sp. 1889 *Shuttleworthia* sp. 1890 *Simonsiella muelleri* 1891 *Slackia equolifaciens*

1892 *Slackia exigua* 1893 *Slackia faecialis* 1894 *Slackia heliotrinireducens* 1895 *Slackia isoflavoniconvertens* 1896 *Slackia piriformis* 1897 *Slackia* sp. 1898 *Solobacterium moorei* 1899 *Sphingobacterium faecium* 1900 *Sphingobacterium mizutaii* 1901 *Sphingobacterium multivorum* 1902 *Sphingobacterium spiritivorum* 1903 *Sphingomonas echinoides* 1904 *Sphingomonas* sp. 1905 *Sphingomonas* sp. 1906 *Sphingomonas* sp. 1907 *Sphingomonas* sp. 1908 *Sphingopyxis alaskensis* 1909 *Spiroplasma insolitum* 1910 *Sporobacter termitidis* 1911 *Sporolactobacillus inulinus* 1912 *Sporolactobacillus nakayamae* 1913 *Sporosarcina newyorkensis* 1914 *Sporosarcina* sp. 1915 *Staphylococcaceae bacterium* 1916 *Staphylococcus aureus* 1917 *Staphylococcus auricularis* 1918 *Staphylococcus capitis* 1919 *Staphylococcus caprae* 1920 *Staphylococcus carnosus* 1921 *Staphylococcus cohnii* 1922 *Staphylococcus condimentii* 1923 *Staphylococcus epidermidis* 1924 *Staphylococcus equorum* 1925 *Staphylococcus fleurettii* 1926 *Staphylococcus haemolyticus* 1927 *Staphylococcus hominis* 1928 *Staphylococcus lugdunensis* 1929 *Staphylococcus pasteurii* 1930 *Staphylococcus pseudintermedius* 1931 *Staphylococcus saccharolyticus* 1932 *Staphylococcus saprophyticus* 1933 *Staphylococcus sciuri* 1934 *Staphylococcus* sp. 1935 *Staphylococcus* sp. 1936 *Staphylococcus* sp. 1937 *Staphylococcus succinus* 1938 *Staphylococcus vitulinus* 1939 *Staphylococcus warneri* 1940 *Staphylococcus xylosus* 1941 *Stenotrophomonas maltophilia* 1942 *Stenotrophomonas* sp. 1943 *Streptobacillus moniliformis* 1944 *Streptococcus agalactiae* 1945 *Streptococcus alactolyticus* 1946 *Streptococcus anginosus* 1947 *Streptococcus australis* 1948 *Streptococcus bovis* 1949 *Streptococcus canis* 1950 *Streptococcus constellatus* 1951 *Streptococcus cristatus* 1952 *Streptococcus downei* 1953 *Streptococcus dysgalactiae* 1954 *Streptococcus equi* 1955 *Streptococcus equinus* 1956 *Streptococcus gallolyticus* 1957 *Streptococcus genomo* sp. 1958 *Streptococcus genomo* sp. 1959 *Streptococcus genomo* sp. 1960 *Streptococcus genomo* sp. 1961 *Streptococcus genomo* sp. 1962 *Streptococcus genomo* sp. 1963 *Streptococcus genomo* sp. 1964 *Streptococcus genomo* sp. 1965 *Streptococcus gordonii* 1966 *Streptococcus infantarius* 1967 *Streptococcus infantis* 1968 *Streptococcus intermedius* 1969 *Streptococcus lutetiensis* 1970 *Streptococcus massiliensis* 1971 *Streptococcus milleri* 1972 *Streptococcus mitis* 1973 *Streptococcus mutans* 1974 *Streptococcus oligofermentans* 1975 *Streptococcus oralis* 1976 *Streptococcus parasanguinis* 1977 *Streptococcus pasteurianus* 1978 *Streptococcus peroris* 1979 *Streptococcus pneumoniae* 1980 *Streptococcus porcinus* 1981 *Streptococcus pseudopneumoniae* 1982 *Streptococcus pseudoporcinus* 1983 *Streptococcus rattii* 1984 *Streptococcus salvarius* 1985 *Streptococcus sanguinis* 1986 *Streptococcus sinensis* 1987 *Streptococcus* sp. 1988 *Streptococcus* sp. 1989 *Streptococcus* sp. 1990 *Streptococcus* sp. 1991 *Streptococcus* sp. 1992 *Streptococcus* sp. 1993 *Streptococcus* sp. 1994 *Streptococcus* sp. 1995 *Streptococcus* sp. 1996 *Streptococcus* sp. 1997 *Streptococcus* sp. 1998 *Streptococcus* sp. 1999 *Streptococcus* sp. 2000 *Streptococcus* sp. 2001 *Streptococcus* sp. 2002 *Streptococcus* sp. 2003 *Streptococcus* sp. 2004 *Streptococcus* sp. 2005 *Streptococcus* sp. 2006 *Streptococcus* sp. 2007 *Streptococcus* sp. 2008 *Streptococcus* sp. 2009 *Streptococcus* sp. 2010 *Streptococcus* sp. 2011 *Streptococcus* sp. 2012 *Streptococcus* sp. 2013 *Streptococcus* sp. 2014 *Streptococcus* sp. 2015 *Streptococcus* sp. 2016 *Streptococcus* sp. 2017 *Streptococcus* sp. 2018 *Streptococcus* sp. 2019 *Streptococcus* sp. 2020 *Streptococcus* sp. 2021 *Streptococcus* sp. 2022 *Streptococcus* sp. 2023 *Streptococcus* sp. 2024 *Streptococcus* sp. 2025 *Streptococcus* sp. 2026 *Streptococcus* sp. 2027 *Streptococcus* sp. 2028 *Streptococcus* sp. 2029 *Streptococcus* sp. 2030 *Streptococcus* sp. 2031 *Streptococcus* sp. 2032 *Streptococcus* sp. 2033 *Streptococcus* sp. 2034 *Streptococcus* sp. 2035 *Streptococcus* sp. 2036 *Streptococcus* sp. 2037 *Streptococcus* sp. 2038 *Streptococcus* sp. 2039 *Streptococcus* sp. 2040 *Streptococcus suis* 2041 *Streptococcus thermophilus* 2042 *Streptococcus uberis* 2043 *Streptococcus urinalis* 2044 *Streptococcus vestibularis* 2045 *Streptococcus viridans* 2046 *Streptomyces albus* 2047 *Streptomyces griseus* 2048 *Streptomyces* sp. 2049 *Streptomyces* sp. 2050 *Streptomyces* sp. 2051 *Streptomyces* sp. 2052 *Streptomyces* sp. 2053 *Streptomyces thermoviolaceus* 2054 *Subdohgranulum variabile* 2055 *Succinatimonas hippei* 2056 *Sutterella morbirenis* 2057 *Sutterella parvirubra* 2058 *Sutterella sanguinis* 2059 *Sutterella* sp. 2060 *Sutterella stercoricanis* 2061 *Sutterella*

wadsworthensis 2062 Synergistes genomu sp. 2063 Synergistes sp. 2064 Synergistetes bacterium
 2065 Synergistetes bacterium 2066 Synergistetes bacterium 2067 Synergistetes bacterium 2068
 Synergistetes bacterium 2069 Syntrophococcus sucromutans 2070 Syntrophomonadaceae genomu
 sp. 2071 Tannerella forsythia 2072 Tannerella sp. 2073 Tatlockia micdadei 2074 Tatumella ptyseos
 2075 Tessaracoccus sp. 2076 Tetrigenococcus halophilus 2077 Tetrigenococcus koreensis 2078
 Thermoanaerobacter pseudethanolicus 2079 Thermobifida fusca 2080 Thermofilum pendens 2081
 Thermus aquaticus 2082 Tissierella praeacuta 2083 Trabulsiella guamensis 2084 Treponema
 genomu sp. 2085 Treponema genomu sp. 2086 Treponema genomu sp. 2087 Treponema genomu sp.
 2088 Treponema phagedenis 2089 Treponema sp. 2090 Treponema sp. 2091 Treponema sp. 2092
 Treponema sp. 2093 Treponema sp. 2094 Treponema sp. 2095 Treponema sp. 2096 Treponema sp.
 2097 Treponema sp. 2098 Treponema sp. 2099 Treponema sp. 2100 Treponema sp. 2101
 Treponema sp. 2102 Treponema sp. 2103 Treponema sp. 2104 Treponema sp. 2105 Treponema sp.
 2106 Treponema sp. 2107 Treponema sp. 2108 Treponema sp. 2109 Treponema sp. 2110
 Treponema sp. 2111 Tropheryma whipplei 2112 Trueperella pyogenes 2113 Tsukamurella
 paurometabola 2114 Tsukamurella tyrosinosolvens 2115 Turicibacter sanguinis 2116 Ureaplasma
 parvum 2117 Ureaplasma urealyticum 2118 Ureibacillus composti 2119 Ureibacillus suwonensis
 2120 Ureibacillus terrenus 2121 Ureibacillus thermophilus 2122 Ureibacillus thermosphaericus
 2123 Vagococcus fluvialis 2124 Veillonella atypica 2125 Veillonella dispar 2126 Veillonella
 genomu sp. 2127 Veillonella montpellierensis 2128 Veillonella parvula 2129 Veillonella sp. 2130
 Veillonella sp. 2131 Veillonella sp. 2132 Veillonella sp. 2133 Veillonella sp. 2134 Veillonella sp.
 2135 Veillonella sp. 2136 Veillonella sp. 2137 Veillonella sp. 2138 Veillonella sp. 2139 Veillonella
 sp. 2140 Veillonella sp. 2141 Veillonella sp. 2142 Veillonella sp. 2143 Veillonella sp. 2144
 Veillonellaceae bacterium 2145 Veillonellaceae bacterium 2146 Victivallaceae bacterium 2147
 Victivallis vadensis 2148 Virgibacillus proomii 2149 Weissella beninensis 2150 Weissella cibaria
 2151 Weissella confusa 2152 Weissella hellenica 2153 Weissella kandleri 2154 Weissella koreensis
 2155 Weissella paramesenteroides 2156 Weissella sp. 2157 Wolinella succinogenes 2158
 Xanthomonadaceae bacterium 2159 Xanthomonas campestris 2160 Xanthomonas sp. 2161
 Xenophilus aerolatus 2162 Yokenella regensburgei 2163 Zimmermannella bifida 2164 Zymomonas
 mobilis 2165 Alistipes shahii 2166 Bacteroides caccae 2167 Bacteroides eggerthii 2168
 Bacteroides sp. 2169 Bacteroides sp. 2170 Bacteroides sp. 2171 Bacteroides uniformis 2172
 Bacteroides vulgatus 2173 Bacteroides vulgatus 2174 Bifidobacterium adolescentis 2175
 Bifidobacterium pseudocatenulatum 2176 Blautia producta 2177 Blautia producta 2178 Blautia
 schinkii 2179 Clostridium bolteae 2180 Clostridium butyricum 2181 Clostridium disporicum 2182
 Clostridium hathewayi 2183 Clostridium hylemonae 2184 Clostridium innocuum 2185 Clostridium
 innocuum 2186 Clostridium mayombeii 2187 Clostridium nexile 2188 Clostridium orbiscindens
 2189 Clostridium symbiosum 2190 Clostridium tertium 2191 Collinsella aerofaciens 2192
 Coprobacillus sp. 2193 Coprococcus catus 2194 Coprococcus comes 2195 Dorea formicigenerans
 2196 Dorea longicatena 2197 Enterococcus faecalis 2198 Erysipelotrichaceae bacterium 2199
 Escherichia coli 2200 Escherichia coli 2201 Eubacterium eligens 2202 Eubacterium rectale 2203
 Eubacterium rectale 2204 Faecalibacterium prausnitzii 2205 Faecalibacterium prausnitzii 2206
 Lachnospiraceae bacterium 2207 Odoribacter splanchnicus 2208 Odoribacter splanchnicus 2209
 Parabacteroides merdae 2210 Roseburia intestinalis 2211 Ruminococcus bromii 2212
 Ruminococcus gnavus 2213 Ruminococcus obeum 2214 Ruminococcus torques 2215 Streptococcus
 thermophilus 2216 Escherichia coli 2217 Streptococcus thermophilus 2218 Escherichia coli 2219
 Therapeutic Compositions

(88) Any of the GABA-producing bacteria described herein (e.g., natural bacteria or engineered
 bacteria), or any combination thereof (including combinations of natural and engineered bacteria)
 can be incorporated into a therapeutic composition. For instance, the therapeutic compositions can
 be administered to a patient in need thereof to treat or alleviate the symptom of a mental illness or
 central nervous system disease.

(89) Purification of Strains

(90) In some embodiments, bacteria are purified prior to incorporation into a therapeutic composition. For instance, bacteria can be purified so that the population of bacteria is substantially free of other bacteria (e.g., contains at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, or at least 98%, at least 99% of the specific bacterial strain or strains desired in the composition).

(91) In some embodiments, the therapeutic composition is a probiotic or a medical food comprising at least one GABA-producing bacterial strain. The strain can be administered, for instance, as a probiotic, as capsules, tablets, caplets, pills, troches, lozenges, powders, and/or granules. The strain can also be formulated as a medical food. The GABA-producing bacteria can also be administered as a fecal transplant or suppository.

(92) In some embodiments, the dose of the therapeutic can contain 1×10^4 , 1×10^5 , 1×10^6 , 1×10^7 , 1×10^8 , 1×10^9 , 1×10^{10} , 1×10^{11} or greater than 1×10^{11} colony forming units (CFUs) of the desired bacterial species. For instance, the desired bacterial species can be GABA-producing bacteria, bacteria that are capable of inhibiting the growth of GABA-consuming bacteria, or a combination thereof.

(93) In some embodiments, the therapeutic composition or dose unit comprises a pharmaceutically acceptable formulation, including an enteric coating or similar to survive the acidity of the stomach and enabled delivery into the small or large intestines, prebiotics (such as, but not limited to, amino acids (including arginine, glutamate, and ornithine), biotin, fructooligosaccharide, galactooligosaccharides, hemicelluloses (e.g., arabinoxylan, xylan, xyloglucan, and glucomannan), inulin, chitin, lactulose, mannan oligosaccharides, oligofructose-enriched inulin, gums (e.g., guar gum, gum arabic and carregenaan), oligofructose, oligodextrose, tagatose, resistant maltodextrins (e.g., resistant starch), trans-galactooligosaccharide, pectins (e.g., xylogalactouronan, citrus pectin, apple pectin, and rhamnogalacturonan-I), dietary fibers (e.g., soy fiber, sugarbeet fiber, pea fiber, corn bran, and oat fiber) and xylooligosaccharides, polyamines (such as but not limited to spermidine and putrescine), an effective amount of an anti-bacterial agent, anti-fungal agent, anti-viral agent, or anti-parasitic agent, or any combinations of the above. For instance, the therapeutic composition can also be in the form of a yogurt containing one or more purified strains of GABA-producing bacteria.

(94) Disease Indications

(95) In one or more embodiments of any of the above-aspects, the mental illness or disease of the central nervous system that can be treated by administration of a therapeutic composition described herein is selected from depression, bipolar disorder, schizophrenia, anxiety, anxiety disorders, addiction, social phobia, major depressive disorder, treatment-resistant major depressive disorder (TR-MDD), major depressive disorder and its subtypes (melancholic depression, atypical depression, catatonic depression, postpartum depression, and seasonal affective disorder), Neurodegenerative amyloid disorders (Parkinson's, Alzheimer's, and Huntington's diseases) orthostatic tremor, Lafora disease, restless leg syndrome, neuropathic pain, pain disorders, dementia, epilepsy, stiff-person syndrome, premenstrual dysphoric disorder, autism spectrum disorder, sleep disorders, and attention deficit hyperactivity disorder (ADHD).

(96) In some embodiments, the method further comprises decreasing at least one symptom of a mental disorder or disease of the central nervous system in the subject selected from the group consisting of fatigue, insomnia, motor dysfunction, stress, persistent anxiety, persistent sadness, social withdrawal, substance withdrawal, irritability, thoughts of suicide, thoughts of self-harm, restlessness, low sex drive, lack of focus, loss of appetite, seizures, memory loss, anger, bouts of emotional reactivity, confusion, pain, and muscle spasms.

(97) Methods of Treatment

(98) The therapeutic compositions described herein can be administered to a patient in need thereof, for instance for the treatment of a mental illness or disease of the central nervous system.

In some embodiments, the method of treatment can comprise first diagnosing a patient who can benefit from treatment by a therapeutic composition described herein. In some embodiments, the method further comprises administering to the patient a therapeutic composition described herein.

(99) Patient Diagnosis

(100) In some embodiments, the process of identifying a subject with a mental illness or disease of the central nervous system can be carried out by a trained psychologist, psychiatrist, or neurologist. For instance, a psychiatrist, psychologist, or neurologist can diagnose a subject with a mental illness or disease of the central nervous system evaluating the subject's behavior for symptoms of the mental illness or disease of the central nervous system. One of skill in the art will understand that mental illness can also be identified in a subject with the aid of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5), (American Psychiatric Association).

(101) In one or more embodiments, the process of identifying a subject with a mental illness or disease of the central nervous system can comprise diagnosing the subject with a mental illness or disease of the central nervous system. In some embodiments, the mental illness or disease of the central nervous system is identified or diagnosed using fMRI. In some embodiments, mental illness or disease of the central nervous system can be identified with standard psychological and neurological surveys, or in other methods known to experts in the field.

(102) In some embodiments, a subject in need of treatment with a therapeutic composition described herein can be identified by identifying low levels of GABA in the subject's blood, serum, stool, or other bodily fluid. In some embodiments, the amount of GABA in the subject's stool (e.g., the initial amount of GABA in the subject's stool) is below about 8 μg GABA per gram of stool. The amount of GABA can be measured using the wet or dry weight of stool by LC/MS or another technique known in the art. In some embodiments, the amount of GABA in the subject's blood or serum (e.g., the initial amount of GABA in the subject's blood or serum) is below about 10 $\mu\text{g/L}$ $\pm$ 5 $\mu\text{g/L}$ GABA per gram of blood or serum (e.g., as measured by LC/MS). In some embodiments, the amount of GABA in the prefrontal cortex, or other areas of the brain, is below about 1.0 mM/kg, as measured by proton magnetic resonance (PMR), or another similar technique.

(103) In some embodiments, the percentage of GABA-producing bacteria in the subject's gut (e.g., the initial amount) represents about 10% of total 16S sequences as measured by sequencing using such methods as 16S rDNA gene Illumina sequencing or quantitative PCR. In some embodiments, the percentage of GABA-producing bacteria in the subject's gut represents about 9%, about 8%, about 7%, about 6%, about 5%, about 4%, about 3%, about 2%, about 1%, or less than about 1% of the total 16S sequences measured in the subject's gut.

(104) Determination of the initial amount of GABA in a subject's blood, serum, regions of the brain, or stool can help identify subjects that can benefit from treatment by administration of GABA-producing bacteria. In some embodiments, a subject with an initial amount of GABA in the serum or blood below 10 $\mu\text{g/L}$ GABA can benefit from administration of GABA-producing bacteria. In some embodiments, a subject with an initial amount of GABA in the serum or blood below 100 μg , below 50 μg , below 25 μg , below 20 μg , below 15 μg , below 10 μg , below 9 μg , below 8 μg , below 7 μg , below 6 μg , below 5 μg , below 4 μg , below 3 μg , below 2 μg , below 1 μg , below 0.5 μg , below 0.1 μg , below 0.01 μg , below 10 ng, or below 1 ng, or below 0.1 ng per L of blood or serum can benefit from administration of a GABA-producing bacteria.

(105) In some embodiments, a subject with an initial amount of GABA in the brain, in regions such as the prefrontal cortex (or other areas of the brain), of about 1.0 mM/kg can benefit from treatment by administration of GABA-producing bacteria. In some embodiments, a subject with an initial amount of GABA in the brain, in such regions as the prefrontal cortex (or other areas of the brain), of below 100 mM, below 50 mM, below 25 mM, below 20 mM, below 15 mM, below 10 mM, below 9 mM, below 8 mM, below 7 mM, below 6 mM, below 5 mM, below 4 mM, below 3 mM, below 2 mM, below 1 mM, below 0.5 mM, below 0.1 mM, or below 0.01 mM, or below 0.001 mM GABA can benefit from treatment by administration of GABA-producing bacteria.

(106) In some embodiments, a subject with an initial amount of GABA in stool below 8 μg GABA per gram of stool (wet or dry weight) can benefit from administration of GABA-producing bacteria. In some embodiments, a subject with an initial amount of GABA in the stool below 100 μg , below 50 μg , below 25 μg , below 20 μg , below 15 μg , below 10 μg , below 9 μg , below 8 μg , below 7 μg , below 6 μg , below 5 μg , below 4 μg , below 3 μg , below 2 μg , below 1 μg , below 0.5 μg , below 0.1 μg , below 0.01 μg , below 10 ng, or below 1 ng, or below 0.1 ng per gram of stool can benefit from administration of a GABA-producing bacteria.

(107) In some embodiments of any of the above aspects, the amount of GABA is increased 0.1, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1000, 2000, 3000, 4000, 5000, or more percent in the subject's stool relative to the initial amount of GABA in the subject's stool, e.g., as measured in step (b) of any of the above-aspects. In some embodiments, the amount of GABA is increased 0.1, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1000, 2000, 3000, 4000, 5000, or more percent in the subject's blood or serum relative to the initial amount of GABA in the subject's blood or serum, e.g., as measured in step (b) of any of the above-aspects. In some embodiments, the amount of GABA is increased 0.1, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1000, 2000, 3000, 4000, 5000, or more percent in regions of the subject's brain, such as, but not limited to the prefrontal cortex, relative to the initial amount of GABA in the subject's brain, e.g., as measured in step (b) of any of the above-aspects. In some embodiments, at least one GABA-producing bacteria is increased 0.1, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1000, 2000, 3000, 4000, 5000, or more percent in the subject's stool relative to the initial amount of GABA-producing bacteria in the subject's stool, e.g., as measured in step (b) in any of the above-aspects. In some embodiments, the level of expression of at least one GABA producing enzyme is increased 0.1, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1000, 2000, 3000, 4000, 5000, or more percent in the subject's stool relative to the initial level of expression of GABA-producing enzymes in the subject's stool, as measured by qPCR or some other appropriate method, known to those familiar in the field.

(108) In some embodiments of the disclosure, the amount of GABA-consuming bacteria can be reduced, e.g., reduced in the subject's stool, blood serum, and the like. The GABA-consuming bacteria can be, for instance, *Evtepia gabavorous* or Firmicutes bacterium MGS:114. In some embodiments, GABA-consuming bacteria can be reduced by 0.1, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1000, 2000, 3000, 4000, 5000, or more percent.

(109) Accordingly, the present disclosure provides for the treatment of mental illness or disease of the central nervous system comprising administering to the subject GABA-producing bacteria, or prebiotics to stimulate growth or GABA production capabilities of GABA producing bacteria.

(110) Methods of Culturing GABA-Consuming Bacteria

(111) In some embodiments, the present disclosure provides a method of culturing bacteria that require GABA for survival and replication. In some cases, these bacteria were previously uncultured or unculturable. In some cases, the bacteria are cultured by supplying endogenous GABA to the growth medium. In some embodiments, the bacteria are cultured by co-culturing the bacteria with a different bacterium capable of producing GABA (e.g., a bacterium as described above).

(112) In some embodiments, the previously uncultured bacterium is *E. gabavorous*. *E. gabavorous* can be cultured on a suitable substrate such as agar. In some embodiments, the agar can contain added GABA. In some embodiments, the present disclosure provides a method of culturing *E. gabavorous* comprising co-culturing *E. gabavorous* with another bacterial strain, said strain is capable of producing GABA, for instance at conditions that are physiologically relevant and found in to the human gastrointestinal tract (e.g., pH less between about 4.5 and about 7.5).

(113) Without wishing to be bound by theory, some previously unculturable bacteria (e.g., *E. gabavorous*) may be able to grow in proximity to cultivable organisms producing growth factors necessary for bacteria to survive or grow. Accordingly, the present disclosure teaches the discovery and culture of *E. gabavorous* in the presence of GABA as a necessary growth factor.

(114) *E. gabavorous* was identified as a late-growing colony in spatial proximity to *Bacteroides fragilis* KLE1758. It was found that growth of *E. gabavorous* KLE1738 was induced in the presence of supernatant derived from *Bacteroides fragilis* KLE1758. Chemical analysis via HPLC and NMR of *Bacteroides fragilis* KLE1738 supernatant revealed GABA as the necessary growth factor for *E. gabavorous*.

(115) As set forth in FIGS. 1A-B, and Example 2, *E. gabavorous* was initially found because it grew in the presence of *Bacteroides fragilis* KLE1758. It was proposed that *Bacteroides fragilis* KLE1758 produces a growth factor that is necessary for the growth and survival of *E. gabavorous*. FIG. 1A shows a photograph of an agar plate containing colonies of bacteria after treatment with human stool. The inset at the top right shows a close-up of a colony of KLE1758 along with a colony of 1738 that is growing in the immediate proximity. FIG. 1B shows a colony of KLE1758 that is capable of supporting multiple colonies of KLE1738 on an agar plate with no other bacteria. Without wishing to be bound by theory, *Bacteroides fragilis* KLE1758 can support the growth of *E. gabavorous* KLE1738. Without wishing to be bound by theory, *Bacteroides fragilis* KLE1758 and *E. gabavorous* KLE1738 can co-exist in a symbiotic relationship in which *E. gabavorous* KLE1738 can consume the GABA that is produced by *Bacteroides fragilis* KLE1758.

(116) As shown in FIGS. 2A-G and Examples 3-4, the supernatant from a 48-hour culture of *Bacteroides fragilis* KLE1758 was found to support the growth of *E. gabavorous* KLE1738, whereas standard agar was not. After a series of purification and isolation steps of the KLE1758 supernatant, it was discovered that GABA was responsible for the growth of *E. gabavorous* KLE1738. FIG. 2A shows that *E. gabavorous* KLE1738 grew in the presence of supernatant of *Bacteroides fragilis*. However, FIG. 2B shows that *E. gabavorous* KLE1738 did not grow in the presence of sterile vehicle on standard agar. After a first fractionation of the *Bacteroides fragilis* KLE1758 spent medium, it was found that the most polar fragment was capable of inducing the growth of *E. gabavorous* KLE1738 (FIG. 2C), but that less polar fragments could not induce growth (FIG. 2D). FIGS. 2E and 2F show close-up views demonstrating that the most polar fraction of the *Bacteroides fragilis* KLE1758 supernatant could induce growth of *E. gabavorous* KLE1738 (FIG. 2E), whereas less polar fragments could not (FIG. 2F). As shown in FIG. 2G, only GABA was identified as being capable of inducing growth of *E. gabavorous* KLE1738.

(117) The 16S nucleotide sequence of *E. gabavorous* KLE1738 is given in Seq. ID No. 2286.

(118) The genetic sequence of *E. gabavorous* KLE1738 was identified as set forth in Example 5. The annotated genome (2,500,009 bp) of *E. gabavorous* is given in the attached Sequence Listing comprising SEQ. ID Nos. 1-2288 and is given in SEQ ID Nos. 2218-2285. Without wishing to be bound by theory, the genome revealed no obvious entry points for metabolism of common sugars or other carbon sources.

(119) Without wishing to be bound by theory, it was discovered that transport systems for common sugars or other carbon sources were also incomplete. Without wishing to be bound by theory, their absence suggests a recent loss of function. *E. gabavorous* is predicted to have a limited set of transporters, including those for methionine, branched-chain amino acids, dipeptides, oligopeptides, and choline/betaine, as predicted in Table 10.

(120) TABLE-US-00011 TABLE 10 Predicted Transport Systems in *E. gabavorous*. Category Subsystem Role Amino Acids and Derivatives Polyamine Metabolism ABC transporter, periplasmic spermidine putrescine-binding protein PotD (TC 3.A.1.11.1) Amino Acids and Derivatives Polyamine Metabolism Spermidine Putrescine ABC transporter permease component PotB (TC 3.A.1.11.1) Amino Acids and Derivatives Polyamine Metabolism Spermidine Putrescine ABC transporter permease component PotC (TC 3.A.1.11.1) Amino Acids and Derivatives Methionine Biosynthesis Methionine ABC transporter ATP-binding protein Amino Acids and Derivatives Methionine Biosynthesis Methionine ABC transporter permease protein Amino Acids and Derivatives Methionine Biosynthesis Methionine ABC transporter substrate-binding protein Amino Acids and Derivatives Methionine Degradation Methionine ABC transporter ATP-binding

protein Amino Acids and Derivatives Methionine Degradation Methionine ABC transporter permease protein Amino Acids and Derivatives Methionine Degradation Methionine ABC transporter substrate-binding protein Clustering-based PhoR-PhoB two-component Phosphate ABC transporter, periplasmic subsystems regulatory system phosphate-binding protein PstS (TC 3.A.1.7.1) Membrane Transport ABC transporter dipeptide (TC Dipeptide-binding ABC transporter, periplasmic 3.A.1.5.2) substrate-binding component (TC 3.A.1.5.2) Membrane Transport ABC transporter oligopeptide (TC Oligopeptide ABC transporter, periplasmic 3.A.1.5.1) oligopeptide-binding protein OppA (TC 3.A.1.5.1) Membrane Transport ABC transporter branched-chain Branched-chain amino acid ABC transporter, amino acid (TC 3.A.1.4.1) amino acid-binding protein (TC 3.A.1.4.1) Phosphorus Metabolism High affinity phosphate transporter Phosphate ABC transporter, periplasmic and control of PHO regulon phosphate-binding protein PstS (TC 3.A.1.7.1) Phosphorus Metabolism Phosphorus Metabolism Phosphate ABC transporter, periplasmic phosphate-binding protein PstS (TC 3.A.1.7.1) Stress Response Choline and Betaine Uptake and L-proline glycine betaine ABC transport system Betaine Biosynthesis permease protein ProV (TC 3.A.1.12.1) Stress Response Choline and Betaine Uptake and Glycine betaine ABC transport system, glycine- Betaine Biosynthesis betaine-binding protein OpuAC Stress Response Choline and Betaine Uptake and Glycine betaine ABC transport system, permease Betaine Biosynthesis protein OpuAB Stress Response Choline and Betaine Uptake and L-proline glycine betaine binding ABC transporter Betaine Biosynthesis protein ProX (TC 3.A.1.12.1) Sulfur Metabolism Alkanesulfonate assimilation ABC-type nitrate/sulfonate/bicarbonate transport system, ATPase component Sulfur Metabolism Alkanesulfonate assimilation Alkanesulfonates ABC transporter ATP-binding protein

(121) Without wishing to be bound by theory, these amino acids are not usually capable of supporting bacterial growth as single carbon sources, unlike serine, threonine, glutamate, and others. This is supported by the inability of *E. gabavorous* to grow on the tested amino acids.

(122) Without wishing to be bound by theory, the metabolic pathway of *E. gabavorous* is proposed to be similar to that of *Clostridium aminobutyricum*, as set forth in FIG. 3, as all enzymes in this pathway were identified in the *E. gabavorous* genome (Table 11).

(123) TABLE-US-00012 TABLE 11 Enzymes in *E. gabavorous* GABA fermentation pathway, predicted by RAST

Enzyme #	Contig	Start	Stop	Length (bp)	Function
1	KLE1738_5	26729	28084	1356	Gamma-aminobutyrate:alpha-ketoglutarate aminotransferase (EC 2.6.1.19)
2	KLE1738_5	25576	26688	1113	NAD-dependent 4-hydroxybutyrate dehydrogenase (EC 1.1.1.61)
3	KLE1738_2	195690	197033	1344	4-hydroxybutyrate:acetyl-CoA CoA transferase (EC 2.8.1.—)
3	KLE1738_28	9343	10665	1323	4-hydroxybutyrate:acetyl-CoA CoA transferase (EC 2.8.1.—)
3	KLE1738_5	24249	25547	1299	4-hydroxybutyrate:acetyl-CoA CoA transferase (EC 2.8.1.—)
3	KLE1738_7	87690	86401	1290	4-hydroxybutyrate:acetyl-CoA CoA transferase (EC 2.8.1.—)
4	KLE1738_5	22352	23902	1551	4-hydroxybutanoyl-CoA dehydratase (EC 4.2.1.—)/Vinylacetyl-CoA Delta-isomerase (EC 5.3.3.3)
5	KLE1738_6	2958	2092	867	Enoyl-CoA hydratase (EC 4.2.1.17)
5	KLE1738_11	1132	356	777	Enoyl-CoA hydratase (EC 4.2.1.17)
6	KLE1738_6	4710	5762	1053	3-hydroxybutyryl-CoA dehydrogenase (EC 1.1.1.35)
6	KLE1738_8	87921	88769	849	3-hydroxybutyryl-CoA dehydrogenase (EC 1.1.1.35)
7	KLE1738_14	4910	6139	1230	Acetyl-CoA acetyltransferase (EC 2.3.1.9)
7	KLE1738_19	28638	27382	1257	Acetyl-CoA acetyltransferase (EC 2.3.1.9)
8	KLE1738_13	26709	25738	972	Phosphate acetyltransferase (EC 2.3.1.8)
9	KLE1738_1	119924	121141	1218	Acetate kinase (EC 2.7.2.1)
1b	KLE1738_30	8650	9999	1350	NADP-specific glutamate dehydrogenase (EC 1.4.1.4)
5b	KLE1738_6	4303	3083	1221	Butyryl-CoA dehydrogenase (EC 1.3.8.1)
6b	KLE1738_1	158124	159767	1644	Acetyl-CoA:acetoacetyl-CoA transferase (EC 2.8.3.8)

(124) The pH dependency of *B. fragilis* KLE1758's ability to produce GABA was investigated. As set forth in Example 6, *B. fragilis* KLE1758 was grown at various pH values, and the supernatant from that growth was analyzed using LCMS. As shown in FIG. 4A, GABA is produced primarily at

relatively lower pH (e.g., about 5.5 and below), compared with glutamate.

(125) Accordingly, it was found that *B. fragilis* KLE1758 can produce GABA at low pH, whereas it was found to produce primarily glutamate at relatively high pH. As set forth in Example 6 and FIG. 4A, at a pH of about 5 and about 5.5, *B. fragilis* KLE1758 produced considerably more GABA than glutamate. However, at a pH of about 6 and about 6.5, *B. fragilis* KLE1758 was found to produce primarily glutamate and relatively low quantities of GABA.

(126) Biological Screens for GABA-Producers

(127) The present disclosure also teaches methods of identifying bacteria that can produce GABA. Given the strict requirement of GABA for the growth of *E. gabavorous*, the present disclosure provides methods of screening for bacteria capable of producing GABA using, for instance, *E. gabavorous* and/or other GABA-dependent bacterial growth as a bioassay. Importantly, by using buffered media (e.g., buffered agar), the assay technique set forth herein can be used to identify bacteria that are capable of producing bacteria at various pH values (e.g., between about 5.5 to about 7.5).

(128) As set forth in Example 7, a sample that is thought to contain GABA-producing bacteria, such as a human stool sample, can be mixed with molten agar. The agar, containing the bacterial sample, can then be streaked with a dilute solution of *E. gabavorous*. As shown herein, *E. gabavorous* cannot grow in the absence of GABA, and therefore any colonies of *E. gabavorous* that do form will necessarily grow in close spatial proximity to GABA-producers.

(129) Because GABA production by some bacteria, including *E. coli*, only occurs at a very low pH (e.g., at a pH not relevant to the human gut), the assay method set forth herein was adapted to control the pH of the media, enabling identification of organisms capable of producing GABA at a pH between about 4.5 and about 7.5. Without wishing to be bound by theory, the pH of between about 4.5 and about 7.5 is the relevant pH within the human gut. Accordingly, bacteria that can produce GABA at these pH values can in some embodiments be capable of producing GABA in the human gut.

(130) In other words, by controlling the pH of the growth medium (e.g., by buffering the molten agar), the present disclosure can allow one to distinguish between GABA-producers that are capable of producing GABA at a physiologically relevant pH (e.g., between about 4.5 to about 7.5) from bacteria that are not able to produce GABA at a physiological relevant pH (e.g., bacteria that can only produce substantial quantities of GABA below pH of about 4.5).

(131) Using this method, a number of representatives from multiple genera, including, but not limited to, *Bacteroides*, *Bifidobacterium*, *Blautia*, *Coprococcus*, *Gordonibacter*, *Dorea*, and *Clostridium* were identified. FIG. 4B shows a representative agar plate showing growth of *E. gabavorous* in the presence of a GABA-producing bacteria. FIG. 4C shows a phylogenetic tree of GABA-producing bacteria that were identified using this method.

(132) FIG. 5 shows the GABA production capabilities of certain strains of GABA producers identified using the techniques described herein. As set forth in FIG. 5, eight strains of GABA producers were grown in buffered media (e.g., between about pH 4.5 and about pH 5.0; and between about pH 6.5 and about pH 7.0). Using the method described in Example 6, the GABA-producing capabilities of the GABA-producing bacteria at various pH values were investigated. As set forth in FIG. 5, certain bacteria (e.g., *B. dorei* KLE1912) produced relatively similar quantities of GABA at lower pH (e.g., between about 4.5 and about 5.0). In contrast, certain bacteria produced different amounts of GABA depending on the pH (e.g., *B. vulgatus* KLE1910 and *B. ovatus* KLE1770). Notably, as shown for *B. vulgatus* KLE1910 and *B. ovatus* KLE1770, some bacteria were found to produce relatively more GABA at lower pH than at higher pH, whereas some bacteria were found to produce relatively more GABA at higher pH than at lower pH.

(133) In some embodiments of the method of identifying bacteria capable of producing GABA, the substrate is agar. In some embodiments, the step of contacting the substrate with *E. gabavorous* comprises streaking the agar with a dilute solution of *E. gabavorous*. The GABA-producing

colonies are then identified by growth induction of *E. gabavorous*. As set forth above, growth *E. gabavorous* is used to determine if a bacterial strain produces GABA. However, one of skill in the art will understand that any bacteria that has a strict requirement of GABA for growth and survival can likewise be used as described above to identify bacteria that can produce GABA.

EXAMPLES

(134) The disclosure is further illustrated by the following examples and synthesis examples, which are not to be construed as limiting this disclosure in scope or spirit to the specific procedures herein described. It is to be understood that the examples are provided to illustrate certain embodiments and that no limitation to the scope of the disclosure is intended thereby. It is to be further understood that resort may be had to various other embodiments, modifications, and equivalents thereof which may suggest themselves to those skilled in the art without departing from the spirit of the present disclosure and/or scope of the appended claims.

(135) Unless otherwise noted, all materials were obtained from commercial suppliers and were used without further purification. Anhydrous solvents were obtained from Sigma-Aldrich (Milwaukee, WI) and used directly.

(136) Unless otherwise specified, PCR was performed using the general bacterial primers 27F (5'-AGAGTTTGATCMTGGCTCAG-3') (set forth in Seq ID No. 2287) and 1492R (5'-TACGGYTACCTTGTTACGACTT-3') (set forth in Seq ID No. 2288) to amplify the 16S rRNA gene. The PCR reaction mixture was 12.5 μ L GoTaq Master Mix (Promega), 1 μ L 10 μ M 27F and 1492R primers, 9.5 μ L Nuclease Free Water (Promega), and 1 μ L of a colony resuspended in 100 μ L sterilized distilled water. The amplification conditions were one cycle of 95° C. for 5 min; 30 cycles of 95° C. for 30 s, 55° C. for 30 s; 72° C. for 90 s; and finally one cycle of 72° C. for 7 min. Amplification of PCR reactions were then confirmed using gel electrophoresis on a 0.8% agarose gel loaded saturated with ethidium bromide. Successful PCRs were sequenced by MacroGen Corporation using the 27F primer using the Applied Biosystems 3730xl DNA analyzer. Quality control for sequences was performed using DNA Baser (www.DnaBaser.com), in which ends were trimmed until there were more than 75% good bases (defined by having a QV score of higher than 25) in an 18 base window. Identification of phylogenetic neighbors and calculation of pairwise sequence similarity were carried out using the EzTaxon server.

Example 1: Human Stool Collection

(137) Stool samples from a healthy human donor were collected using a commercially available stool collection vessel. Within 5 minutes of collection, 1 gram of stool was resuspended in 9 mL of sterile 20% glycerol in PBS and homogenized for 30 seconds using a vortex. 1 mL aliquots of this mixture were loaded in cryotubes and stored at -80° C. for cultivation.

Example 2: Cultivation of Helper-Uncultured Pairs from Human Stool Samples

(138) All cultivation work was performed in a Coy Anaerobic Vinyl chamber with an atmosphere of 5% hydrogen, 10% CO₂, 85% nitrogen. Anaerobically, serial dilutions of thawed stool samples were prepared in PBS and bead spread (7-10 beads/plate) on 1 $\times$ Fastidious Anaerobic Agar (Accumedia) plates with 2.5% yeast extract (FAAy). Plates were incubated at 37° C. anaerobically for one week, and each day appearance of colonies were tracked by spotting the outside of the plates with different colored markers. At the end of the week, serial dilutions of late forming colonies (appearance after 4-7 days) were prepared in PBS and bead spread on FAAY plates. Nearby (<2 cm), early forming colonies (appearance after 1-3 days) were then resuspended in PBS at a high density. Five μ L of this suspension were spotted on the plates with their respective spread-candidate dependent and incubated for up to one week in the chamber, and observed daily. Growth induction of the dependent organism around the spotted helper indicated a positive hit.

(139) A fecal sample from a healthy human donor was diluted and spread-plated on rich medium, and newly formed colonies were noted daily for a week. Late forming colonies (3-7 days) were diluted and spread on a nutrient agar plate, and a heavy inoculum of a neighboring, early forming colony (1-2 days) was spotted, as shown in FIG. 1A. Using this method, a number of helper-

uncultured pairs were identified, in which the spread-plated uncultured isolate formed a gradient of growth around the spotted culturable helper.

(140) One isolate, *E. gabavorous* KLE1738 (93.22% similar to *Flavonifractor plautii* ATCC 29863 by 16S rRNA gene sequence), was dependent on *Bacteroides fragilis* KLE1758 (100% similar to *Bacteroides fragilis* ATCC 25852 by 16S rRNA gene sequence) for growth (FIG. 1B). *E. gabavorous*, is a gram positive organism of the Clostridia class.

Example 3: Determination of GABA as Growth Factor for *E. gabavorous*

(141) The supernatant of a 48-hour culture of *B. fragilis* KLE1758 grown in rich medium induced growth of *E. gabavorous* as shown in FIGS. 2A and 2B, enabling bioassay-driven purification of the growth factor. The supernatant was solvent-partitioned with ethyl acetate, and the water residue fraction induced the growth of *E. gabavorous*. The water fraction was then purified using HP-20 column chromatography, and the most polar fraction induced the growth of *E. gabavorous* as shown in FIG. 2C. This active fraction was then further fractionated by preparative HPLC. HPLC yielded one active fraction, and NMR showed that it contained 10 compounds, primarily GABA, threonine, lactic acid, valine, glutamine, malonic acid, succinic acid, and alanine. All compounds were spotted on plates where *E. gabavorous* was spread, and only GABA caused growth induction, as shown in FIGS. 2E-F. The compounds were identified by NMR analysis including .sup.1H, .sup.13C, .sup.1H-.sup.1H COSY, TOCY, HSQC, and HMBC NMR experiments to identify the constituents in the fraction. All NMR experiments were carried out on a Varian INOVA 600 MHz NMR spectrometer equipped with an indirect detection probe.

Example 4: Testing Other Compounds for Induction of *E. gabavorous*

(142) Multiple compounds were tested for the ability to induce the growth of *E. gabavorous* as shown in FIG. 2G. Stocks of each compound (purchased from Sigma, excluding the ATCC Mineral and Vitamin mixes) were prepared dependent on solubility in water at the concentrations shown in FIG. 2G. Five µL of the stocks were then spotted on FAAY plates spread with *E. gabavorous*, and incubated anaerobically for one week, and any growth was observed. No compounds induced growth but GABA.

Example 5: Whole Genome Sequencing and Annotation

(143) DNA from cells of *E. gabavorous* grown 48 hours anaerobically on FAAY plates with 1.0 mg/mL GABA was isolated for genome sequencing using the PowerSoil® DNA Isolation Kit (Mo Bio, San Diego, CA) to manufacturer specifications, yielding ~5.0 µg of high quality DNA. Genomic sequencing and de novo assembly was performed by the Genomic Core at Tufts University in Boston, MA. The genome of *E. gabavorous* was sequenced on an Illumina MiSeq using MiSeq V2 500 cycles chemistry with a paired-end 250 bases format. Briefly, 100 ng of genomic DNA was sheared on a Covaris M220 to an average fragment size of around 600 bases. Using the fragmented DNA as input, a sequencing library was prepared with Illumina TruSeq Nano DNA Sample Preparation Kit per the manufacturer instruction. Base calling and demultiplexing was performed on the raw data from the MiSeq using CASAVA and fastq files were generated. De novo assembly of the genome was performed using Edena V3.131028 with a customized parameter optimization pipeline. The best assembled genome, as assessed by the contig statistic, was reported. Assembly yielded 68 contigs (n), with all contigs having a sequence length longer than 200 bases (n:200). 7 contigs with a larger value than the N50 (119748), and the minimal contig length is 355 (min). The N20, N50 and N20 are 33403, 119748 and 204670, respectively. The largest contig length (max) was 344080, and the estimated genome size is 2500009. The draft genome was annotated using the RAST server and the KAAS (KEGG Automatic Annotation Server) analysis tool of the KEGG (Kyoto Encyclopedia of Genes and Genomes) database. The genome of *E. gabavorous* was annotated using RAST, and the genomes of CAG:113 and *E. gabavorous* were compared using RAST.

Example 6: Quantification of Glutamate and GABA Production in *B. fragilis*

(144) The absolute amount of glutamate and GABA contained in the *B. fragilis* KLE1758

supernatant was determined by HPLC, using a fluorophore to aid in detection. Specifically, free amines were labeled for analysis by reacting with the AccQ reagent (Waters) according to manufacturer's protocols. A calibration curve was generated from stock solutions (10 mg/mL) that were prepared by dissolving GABA (2.0 mg) in water (200 μ L), glutamate (Glu) (10.2 mg) in water (1020 μ L) and cysteic acid (CSA) (16.9 mg) in water (1690 μ L). These were serially diluted to generate a concentration gradient. Specifically, stocks were made to final concentrations of 0.1 mg/mL, 0.05 mg/mL, 0.01 mg/mL and 0.001 mg/mL. An aliquot of a given stock solution, was added to the AccQ reaction buffer (25 μ L final), followed by the addition of the acetonitrile-dissolved AccQ reagent (25 μ L). This was reacted for ten minutes at 55-60° C., and then transferred directly to an LCMS vial, fitted with a glass insert. Reaction concentrations of the amino acids were: 20 ng/ μ L, 10 ng/ μ L, 5 ng/ μ L, 2.5 ng/ μ L, 1 ng/ μ L, 0.5 ng/ μ L, 1 ng/ μ L and 0 ng/ μ L (control). These samples were injected (10 μ L) on an Agilent LCMS, using a gradient of solvent A (water/0.1% formic acid) and B (acetonitrile/0.1% formic acid) over the following time course: 1) 0-40 minutes linear gradient of 2% B to 98% B; 2) 40-45 min isocratic at 98% B; 3) 45-45.5 min linear gradient of 98% B to 2% B; 4) 45.5-55 min isocratic at 2% B. The CSA-AccA derivative eluted at 9.5 min, the Glu-AccQ derivative eluted at 12.1 min and the GABA-AccQ derivative at 12.5 min. The area under the curve, in extracted ion (EIC) mode (m/z =274 for GABA-AccQ, 318 for Glu-AccQ and 340 for CSA-AccQ), was used to develop a calibration curve by plotting area against amount of original Glu, CSA or GABA (in ng) injected. An average of two runs for each tested concentration was used to generate the calibration curve. In the case of GABA and Glu, CSA was added to all reactions, to a final concentration of 2.5 μ g/mL, and used as an internal standard. (145) Triplicate cultures of *B. fragilis* KLE1758 were grown in BHIych anaerobically for 48 hours, the cells centrifuged, and the supernatant was filtered through a 0.2 μ m filter. Samples were stored at 4° C. until analysis. To analyze the samples, an aliquot (2 μ L) of each sample was added to AccQ reaction buffer (16 μ L), CSA internal standard (2 μ L of a 50 μ g/mL solution in buffer), followed by the addition of the AccQ reagent (20 μ L). These samples were heated to 55° C. for ten minutes, and then transferred directly into an LCMS vial fitted with a glass insert. An aliquot of each sample (10 μ L) was injected onto the LCMS, and separated following the same injection program as used for the calibration curve. The total EIC area under curves representing GABA, Glu and CSA was determined using ChemStation software (Agilent). Each injection represented 25% of the original media concentration, therefore the total amount of sample determined (in ng) was multiplied by a factor of four to determine the original concentration (in ng/ μ L= μ g/mL). All areas were normalized to the area under the curve of the internal standard (CSA), which was held at constant concentration throughout the experiment. The results are given in FIG. 4A.

Example 7: Co-Culture Screen for GABA Producers Using *E. gabavorous*

(146) GABA secretion can allow bacteria to survive acid stress. Decarboxylation of glutamate produces GABA, which is exported from the cell in a protonated form, alkalinizing the cytoplasm. *E. coli*, as well as some *Lactobacillus* and *Bifidobacterium* strains were shown to produce GABA, but these organisms are typically found at a low abundance in the human intestinal tract, and in the case of *E. coli*, is dependent on low pH (e.g., about 4.2 and below). *Bacteroides fragilis*, the helper of *E. gabavorous*, is a common gut bacterium, but it was found that similarly to *E. coli*, GABA production by *Bacteroides fragilis* KLE1758 is only observed at a pH less than about 5.5 as shown in FIG. 4A. GABA is shown in left-hand columns and glutamate is shown in right-hand columns at each pH value. Without wishing to be bound by theory, it was therefore considered useful to identify microorganisms capable of producing GABA at a physiologically relevant pH for the human large intestine (e.g., pH of about 5.5 to about 7.5, or about pH 5.7 to about 7.4).

(147) To accomplish this, the strict GABA requirement of *E. gabavorous* was utilized to screen for bacteria capable of secreting GABA on heavily buffered medium. Metabolic byproducts of bacterial growth may lower the pH of the medium in the absence of buffer. Stool sample was mixed with molten agar and poured in Petri plates in an anaerobic chamber, and *E. gabavorous* was spread

on top of the agar once solidified. By looking for zones of growth induction of *E. gabavorous*, and measuring the pH of the agar, bacteria that produce GABA at a pH of between about 6.0 and about 7.0 were identified, as well as those producing GABA at a pH of about 4.5 to about 5.0, as shown in FIG. 4B. The full 16S rRNA gene was amplified and sequenced using the 27F and 1492R universal primers, and annotation with EZTaxon revealed a number of representatives from multiple genera, including *Bacteroides*, *Bifidobacterium*, *Blautia*, *Coprococcus*, *Gordonibacter*, *Dorea*, and *Clostridium* (FIG. 4C). Of these, only *Bifidobacterium adolescentis* was previously reported to produce GABA.

Example 8: Using an Engineered *Escherichia coli* Strain to Produce GABA and Induce the Growth of *E. gabavorous*

(148) GABA can be produced by intestinal epithelial cells and by some bacteria, such as *Escherichia coli* and *Listeria monocytogenes*, by decarboxylation of glutamate. In *E. coli*, the decarboxylation of glutamate serves as a mechanism to decrease intracellular pH, and GABA production generally occurs at a low pH. To survey whether *E. coli* could be engineered to produce GABA, *E. coli* colonies harboring native glutamate decarboxylases (*gadA*, *gadB*), or the GABA antiporter, (*gadC*) in the pCA24N IPTG inducible high-copy number vector, were tested for GABA production via co-cultivation assay with *E. gabavorous*. Overexpression of glutamate decarboxylase in *E. coli* (*gadA* or *gadB*), resulted in induction of KLE1738 growth to levels seen with *B. fragilis*, while expression of the GABA antiporter, *gadC*, did not (FIG. 6). Altering the pH of growth media for KLE1738 did not change the GABA-dependency phenotype. Without wishing to be bound by theory, this suggests that engineering bacteria to overexpress glutamate decarboxylase or other GABA producing enzymes, constitutively or inducibly, is an effective way to produce GABA, as well as induce the growth of *E. gabavorous*.

EQUIVALENTS

(149) While the present disclosure has been described in conjunction with the specific embodiments set forth above, many alternatives, modifications and other variations thereof will be apparent to those of ordinary skill in the art. All such alternatives, modifications and variations are intended to fall within the spirit and scope of the present disclosure.

Claims

1. A method of treating depression in a subject in need thereof, the method comprising administering to the subject a therapeutic composition comprising at least one purified bacterial population that produces GABA at a pH range of between 4.5 and 7.5, the at least one purified bacterial population consisting of bacteria comprising a 16s rDNA sequence at least 98% identical to a 16s rDNA sequence selected from the group consisting of SEQ ID NOs: 1-4, 8, 10, 12, 16-18, 28-29 and 81.
2. The method of claim 1, wherein the at least one bacterial population consists of bacteria comprising a 16S rDNA sequence at least 99% identical to a 16S rDNA sequence selected from the group consisting of SEQ ID NOs: 1-4, 8, 10, 12, 16-18, 28-29 and 81.
3. The method of claim 1, wherein the depression comprises major depressive disorder and its subtypes.
4. The method of claim 3, wherein the subtype of major depressive disorder comprises melancholic depression, atypical depression, catatonic depression, postpartum depression, or seasonal affective disorder.
5. The method of claim 1, wherein the depression comprises treatment-resistant major depressive disorder (TR-MDD).
6. The method of claim 1, wherein treating the depression comprises decreasing at least one symptom of the depression.
7. The method of claim 6, wherein the at least one symptom comprises fatigue, insomnia, stress,

persistent anxiety, persistent sadness, social withdrawal, substance withdrawal, irritability, thoughts of suicide, thoughts of self-harm, restlessness, low sex drive, lack of focus, anger, bouts of emotional reactivity, confusion, loss of appetite, or a combination thereof.

8. The method of claim 1, wherein the at least one purified bacterial population consists of bacteria selected from the group consisting of: *Bacteroides caccae*; *Bacteroides clarus*; *Bacteroides dorei*; *Bacteroides finegoldii*; *Bacteroides stercoris*; *Bacteroides uniformis*; *Bacteroides xylanisolvens*; *Butyricimonas virosa*; *Clostridium perfringens*; *Clostridium sordellii*; *Parabacteroides distasonis*; *Parabacteroides merdae*; *Bacteroides salyersiae*, and combinations thereof.

9. The method of claim 1, wherein the at least one purified bacterial population consists of bacteria comprising a DNA sequence which encodes an enzyme selected from: glutamate decarboxylase; putrescine aminotransferase; gamma-aminobutyraldehyde dehydrogenase; arginine decarboxylase; agmatinase; ornithine decarboxylase; or a combination thereof.

10. The method of claim 9, wherein the glutamate decarboxylase, putrescine aminotransferase, gamma-aminobutyraldehyde dehydrogenase, arginine decarboxylase, agmatinase, ornithine decarboxylase, or a combination thereof, is encoded by a DNA sequence at least 50% identical in DNA sequence to any one of Seq. ID. NOs: 275-304.

11. The method of claim 9, wherein: (a) the glutamate decarboxylase is encoded by a DNA sequence at least 50% identical in DNA sequence to a sequence selected from SEQ ID NOs: 275-279; (b) the putrescine aminotransferase is encoded by a DNA sequence at least 50% identical to a sequence selected from the group consisting of SEQ ID NOs: 280-284; (c) the gamma-aminobutyraldehyde dehydrogenase is encoded by a DNA sequence at least 50% identical to a sequence selected from the group consisting of SEQ ID NOs: 285-289; (d) the arginine decarboxylase is encoded by a DNA sequence at least 50% identical to a sequence selected from the group consisting of SEQ ID NOs: 290-294; (e) the agmatinase is encoded by a DNA sequence at least 50% identical to a sequence selected from the group consisting of SEQ ID NOs: 295-299; or (f) the ornithine decarboxylase is encoded by a DNA sequence at least 50% identical to a sequence selected from the group consisting of SEQ ID NOs: 300-304.

12. The method of claim 1, wherein the composition is in the form of a capsule, a tablet, a caplet, a pill, a troche, a lozenge, a powder, a granule, a medical food, a fecal transplant or a combination thereof.

13. The method of claim 1, wherein the composition further comprises a prebiotic.

14. The method of claim 1, wherein the composition is administered as a fecal transplant.

15. The method of claim 1, further comprising identifying a subject that would benefit from an increase in endogenous GABA by measuring an initial amount of GABA in the subject's stool, blood, serum or brain, wherein the subject is identified as benefitting from an increase in endogenous GABA when: (a) the initial amount of GABA in the subject's stool is below about 8 μ g per gram of wet or dry stool; (b) the initial amount of GABA in the subject's blood or serum is below about 10 μ g per liter of blood; or (c) the initial amount of GABA in the subject's brain is below about 1.0 mM/kg.

16. The method of claim 1, wherein the subject is identified as benefitting from an increase in endogenous GABA when an initial amount of GABA-producing bacteria in the subject's stool is less than about 10% of total bacteria as measured by 16S sequence mapping.
